

STUDENT ASSESSMENT SUBMISSION AND DECLARATION

When submitting evidence for assessment, each student must sign a declaration confirming that the work is their own.

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Issue Date: 16.05.2024	Submission Date: 11.08.2024	Submitted on: 18.08.2024	
Programme: Pearson BTEC HND in Computing			
Unit no. & Title: 04. Database Design and Development			
Assignment title: Database system for “Dickwella Beach Hotel and Resort”			

Plagiarism

Plagiarism is a particular form of cheating. Plagiarism must be avoided at all costs and students who break the rules, however innocently, may be penalised. It is your responsibility to ensure that you understand correct referencing practices. As a university level student, you are expected to use appropriate references throughout and keep carefully detailed notes of all your sources of materials for material you have used in your work, including any material downloaded from the Internet. Please consult the relevant unit lecturer or your course tutor if you need any further advice.

Student Declaration

Student declaration

I certify that the assignment submission is entirely my own work and I fully understand the consequences of plagiarism. I understand that making a false declaration is a form of malpractice.

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Date: **18.08.2024**

Assessment Tracking

Programme:	HND Computing	Student Name:	MN. Sihanass/1030371		
Unit No. & Title:	Unit 4: Database Design and Development	Assessment Issue Date:		Submission Date:	
Assessor Name:	Eng. A.L. Jubailah Begum	Completion Date:		IV Signature:	nismy@bcas.lk

Tasks	Sub tasks	Criteria Targeted	Date Issued	Hand In Date	Formative Feedback	Resubmission Date
01	1.1	P1	16.05. 2024	21.05. 2024	22.05. 2024	25.05. 2024
	1.2	P1, M1				
02	2.1	P2		05.06. 2024	06.06. 2024	10.06. 2024
	2.2	P3				

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	2.3	P3, M2, M3, D2				
03	3	P4, M4		19.06. 2024	20.06. 2024	25.06. 2024
04	4.1	P5, M5, D3		10.07. 2024	11.07. 2024	15.07. 2024
	4.2	M1, D1				

Assignment Brief

Unit 4: Database Design and Development

Student Name & Zoho ID:	MN. Sihanass / 1030371
Programme:	HND Computing
Unit Number and Title	Unit 4: Database Design and Development
Academic Year	2024
Batch	Batch 22
Unit Assessor	Eng. A.L. Jubailah Begum
Assignment Title	Database system for “Dickwella Beach Hotel and Resort”
Issue Date	16.05.2024
Submission Date	21.07.2024
Submission Format	

Pdf document of the report, database log files, SQL files, GUI .exe file and dataedo documentation file.

Unit Learning Outcomes

LO1 Use an appropriate design tool to design a relational database system for a substantial problem

LO2 Develop a fully-functional relational database system, based on an existing system design

LO3 Test the system against user and system requirements

LO4 Produce technical and user documentation.

Transferable skills and competencies developed

Determining user and system requirements, entity relationship modelling, referential integrity, data normalisation to third normal form, database and platform options for system development.

Implementation of the physical data model based on the logical model and linking code to data sets, Data manipulation using appropriate query tools, including complex queries to query across multiple tables and using functions and formulae. Database maintenance and data manipulation: inserts, updates, amendments, deletions, data backup and recovery.

Test procedures to be used: test plans, test models, e.g. white box, black box; testing documentation.

User documentation, including how to use the system, outputs produced by the system, menu operations and other functions.

Vocational scenario

Database System for “Dickwella Beach Hotel and Resort”

The management is proud of new owners of **Dickwella Beach Hotel and Resort** with its 100 en-suite rooms, bar and restaurant. They have asked you to develop a system that will manage the room booking processes, keep details of additional services that guests purchase during their stay and produce the final bill for a stay. **The Manager describes the processing involved as follows:**

- A booking or visit starts with a room request coming in from a potential guest. We take details of the reservation request and search for availability by looking in the desk-diary to see which rooms are available for the required Dates. We have 20 single, 60 double and 20 family rooms that can accommoDate up to 5 persons in each.
- If the required rooms are available we create a record of the reservation and include details of the room allocated to the reservation, start and end Dates and the number of guests staying in the room. At this stage we also check to see if the potential guest has

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stayed with us before or not. If they have, we find their details, check they are still accurate and then add their details to the reservation. If they haven't stayed with us before, we take the Name of the guest making the reservation and their contact details and add these to our records.

- If a guest wants to reserve more than one room we create a separate reservation record for each room reserved.
- In reservation Guest should make payment for reservation.
- When guests arrive at the start of their stay, we find their reservation and personal details, checking that their personal details are still correct and updating them as necessary. We then book them in for their stay, we call this an 'Occupancy'. Local laws require that we record the Names and age of all guests and which room they are staying in.
- We provide a restaurant and theatre booking service for our guests. This process involves little more than a telephone call to the organization and a verbal agreement being reached between the Hotel and the organization.
- During their stay at the Guest House guests may purchase or hire additional services and products from us, such as newspapers, cycles, maps, food and drinks. We need to record all of these so that we can Charge for them at the end of the stay.
- At the end of their stay guests request their final bill. We would like the system to calculate and produce this final bill. Reservation payment will deduct from the full payment. We then give it to the guest so that payment can be made. We keep a record of how much each stay in a particular room was billed for so that we can calculate income per room over a given period of time.
- Guest can cancel the reservation before the reservation start Date. In cancellation the payment for reservation will not be refunded.
- For each room a room boy and waiter are assigned. Room boys and waiters can provide services for many rooms.
- In the event of Guest dissatisfactions about the rooms and services, they can complain to the manager of this hotel. This complains are recorded for further actions.

Assume you are appointed as a Database Administrator for **Dickwella Beach Hotel and Resort** and it is your responsibility is to design a database system for **Dickwella Beach Hotel and Resort**.

Assignment activity and guidance

Task 1

Task 1.1

Analyse and **design a relational database system** for **Dickwella Beach Hotel and Resort** with clear explanations and indicate the designing steps (Conceptual, Logical, and Relational Mapping). Make Any Valid Assumptions.

Use Microsoft Visio tool or any other tool to visualize the ER Model.

Task 1.2

- a. Create **at least four interrelated tables** for the above designed ER diagram, **with clear statements of user and system requirements.**

Give at least five rows for each table. Data entries to the tables can be your own assumed values. Evaluate the **data validations** on the data that you insert.

- b. Apply the **Normalization rules** on the above created tables up to third normal form. Mention each step to be followed along with the rules. The primary keys and the foreign keys must be indicated clearly.

Task 2

Task 2.1

Develop the physical relational database by creating the tables for the above designed ER Model with MSSQL Server. The tables created for above task can be utilized.

Identify and enforce all necessary constraints on the tables which include primary key, foreign key constraints and domain constraints etc.

Task 2.2

Implement the query language to insert data to the above created tables. So as to show your knowledge in SQL apply the queries to update and delete.

Task 2.3

- a. **Implement complex queries** in the above created tables to retrieve meaningful data and make the data entry very easy for the users.
Note: use the comment making Facility in the MSSQL Server when querying to indicate your step by step process clearly.
- b. Make sure that you create a database system **which includes system security and database maintenance**.
- c. **Assess whether meaningful data has been extracted through the use of query tools to produce appropriate management information.**
- d. **Evaluate the effectiveness of the database solution in relation to user and system requirements and suggest improvements.**

Task 3

Task 3

Critically review and **test the developed database system against user and system requirements**. Address your findings, errors and problems encountered in implementing the database **including test plan and the possible test cases**.

Note: Test procedures to be used: test plans, test models e.g. white box, black box; testing documentation.

Task 4

Task 4.1

- a. **Produce a technical user documentation** for developed database system explaining how to use the database system.

Note: include a **flow chart** showing movement of data through the system and **data flow diagram** describing how the system works.

- b. State the **future improvements** that can be undertaken to **ensure the continued effectiveness of the database system**.

Task 4.2

- a. Produce a **comprehensive design for the fully-functional database system**, which includes user interface design.
- b. Evaluate the reasons that the user interface **design** will meet the **requirements as given effectively**.

Recommended Resources

Please note that the resources listed are examples for you to use as a starting point in your research – the list is not definitive.

- Churcher, C. (2012) Beginning Database Design: From Novice to Professional. 2nd edn. Apress.
- Connolly, T. and Begg, C. (2014) Database Systems: A Practical Approach to Design, Implementation and Management. 6th edn. Global Edition. Pearson.
- Flejoles, R. P. (2018) Database Theory and Application. Arcler Press.
- Karwin, B. (2017) SQL Antipatterns: Avoiding the Pitfalls of Database Programming Pragmatic Programmers, LLC.
- Kroemke, D. and Auer, D. (2012) Database Concepts: International Edition. 6th edn. Pearson.

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Use an appropriate design tool to design a relational database system for a substantial problem		
P1 Design a relational database system using appropriate design tools and techniques, containing at least four interrelated tables, with clear statements of user and system requirements.	M1 Produce a comprehensive design for a fully-functional system, which includes interface and output designs, data validations and data normalisation.	D1 Evaluate the effectiveness of the design in relation to user and system requirements.
LO2 Develop a fully-functional relational database system, based on an existing system design		
P2 Develop the database system with evidence of user interface, output and data validations, and querying across multiple tables.	M2 Implement a fully functional database system, which includes system security and database maintenance. M3 Assess whether meaningful data has been extracted through the use of query tools to produce	D2 Evaluate the effectiveness of the database solution in relation to user and system requirements and suggest improvements.

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P3 Implement a query language into the relational database system.	appropriate management information.	
L03 Test the system against user and system requirements		
P4 Test the system against user and system requirements.	M4 Assess the effectiveness of the testing, including an explanation of the choice of test data used.	D2 Evaluate the effectiveness of the database solution in relation to user and system requirements and suggest improvements.
L04 Produce technical and user documentation		
P5 Produce technical and user documentation.	M5 Produce technical and user documentation for a fully-functional system, including data flow diagrams and flowcharts, describing how the system works.	D3 Evaluate the database in terms of improvements needed to ensure the continued effectiveness of the system.

Guidance for Students

Deliverable: Report should be submitted as a **soft copy** of word processed Report.

NOTE:

- You should include the **COVER PAGE, Assessment Declaration form and Unit Review Plan** of this assignment when you submit your final report.
- If submitted after the extended deadline, the assignment will not be accepted whereas you shall be asked to go for a **NEW assignment**.
- Late Submission is not permitted until otherwise recommended by the Assessor /Course Coordinator.
- Plagiarism will be treated as a very **serious academic misconduct**.

Instructions to students:

1. All assignment should comprise of the standard **Front Cover** given. **No other front page will be accepted.**
2. **Report Writing Guidelines:**
 1. Every Assignment should have an **Introduction** and **Conclusion**.
 2. The standard **Table of Contents** should be generated.

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3. All the **Figures, Tables, Diagram** etc. should be numbered.
4. **Main Heading** Font: **Arial**; Size 16
5. **Sub heading**: Font: **Arial**; Size 14
6. **Body text**: Font: **Arial**; Size 11
7. **Paragraph**: 1.5 spacing
8. **Margins: Top: 1" Bottom: 1" Left: 1" Right: 1"**
9. **Header** – include the module Name on the right hand side
10. **Footer** – include the page number on the right hand side
11. All sections should have continuity and pages should be clearly headed.
12. **References** – clear references for all the materials, books, articles, website etc should be given in accordance with Harvard Reference style (Harvard Anglia 2008)

Acknowledgement

I would like to express my sincere gratitude to Eng. A.L. Jubailah Begum Lecturer at British College of Applied Studies Campus, and Mr. Mohamed Ishraque, Head of British College of Applied Studies Campus Kalmunai Branch, for giving me the chance to work on and finish the database design and development assignment. The outstanding support, counsel, and direction I received from Eng. A.L. Jubailah Begum was essential to finishing my assignment successfully. We could not have completed this project without her help and knowledge. I also want to express my gratitude to each and every lecturer in the British College of Applied Studies Campus's HND programme for their invaluable knowledge and assistance with the assignment. Furthermore, I would like to express my gratitude to all of the instructors at British College of Applied Studies Campus who helped me, either directly or indirectly, with my work.

Finally, I would like to express my sincere gratitude to my family and fellow students for their steadfast support throughout the assignment period. Their support and encouragement helped me overcome obstacles and successfully finish this project.

I sincerely appreciate all of your help and contributions.

Abstract

This project involves the design and implementation of a comprehensive database system for Dickwella Beach Hotel and Resort, a newly acquired property featuring 100 en-suite rooms, a bar, and a restaurant. The objective is to optimize the management of room booking processes, maintain detailed records of additional services purchased by guests, and generate final bills for stays. The database system is designed to handle various functionalities including room reservations, guest Occupancy, additional service management, and accurate billing, all while ensuring compliance with local regulations.

The development process begins with a thorough analysis and design phase, which includes the creation of a conceptual, logical, and relational model. The Entity-Relationship (ER) diagram is visualized using tools such as Microsoft Visio, and the database schema is normalized up to the third normal form to ensure data integrity and eliminate redundancy. Clear and detailed explanations of the design steps are provided, along with any necessary assumptions to ensure completeness.

Following the design phase, the implementation involves creating interrelated tables based on the ER diagram in MSSQL Server. These tables are populated with sample data entries to illustrate the database's functionality. Data validation rules are applied to maintain data quality, and essential constraints such as primary keys, foreign keys, and domain constraints are enforced to ensure the relational integrity of the database. SQL queries are utilized to perform data insertion, updates, and deletions, showcasing proficiency in database manipulation.

The developed database system undergoes rigorous testing and review to ensure it meets user and system requirements. A comprehensive testing plan, including white box and black box testing methods, is implemented to identify and Address any errors or issues. Testing documentation details the findings and ensures that the system functions as intended.

Comprehensive technical user documentation is produced, explaining how to use the developed database system. This includes a flowchart depicting the movement of data through the system and a data flow diagram describing system operations. Recommendations for future improvements are provided to ensure the continued effectiveness and scalability of the database system.

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Introduction

In the hospitality industry, an efficient database system is paramount for managing operations and enhancing guest experiences. As the newly appointed Database Administrator for Dickwella Beach Hotel and Resort, I am tasked with designing a comprehensive database system to streamline the hotel's room booking processes, manage additional guest services, and generate accurate billing.

The scenario involves developing a system for a resort with 100 en-suite rooms categorized into single, double, and family rooms. The database system must handle various functionalities, including reservations, Occupancy management, additional services tracking, billing, and guest history maintenance.

Dickwella Beach Hotel and Resort, under new management, requires a robust system to handle the processes. This assessment will detail the database design process, encompassing the requirements analysis, schema design, normalization, implementation strategies, and testing procedures. The goal is to develop a system that not only meets the current operational needs but is also scalable and adaptable for future enhancements.

By effectively managing these processes, the database system will enhance operational efficiency, improve guest satisfaction, and provide valuable insights into the hotel's performance, ensuring Dickwella Beach Hotel and Resort maintains a competitive edge in the hospitality industry.

Task 1

Task 1.1

Design a relational database system for Dickwella Beach Hotel and Resort.

Conceptual design

ENTITY	ATTRIBUTES
Guest	<u>GuestID</u> , Name, Occupancy, DoB, Address, Age, ContactNo (multi valued)
Room	<u>RoomID</u> , Facility, Type, Charge, Capacity, Status
Reservation	<u>ReservationID</u> , StartDate, EndDate, PaymentStatus, NoGuest
Employee	<u>EmployeeID</u> , Name, Role
Additional Service	<u>ServiceID</u> , ServiceName, ServiceCharge
Bill	<u>BillID</u> , TotalAmount, DeductedAmount, FinalAmount
Complaint	ComplaintID, Date, Description

Table 1: Conceptual Design

ER diagram

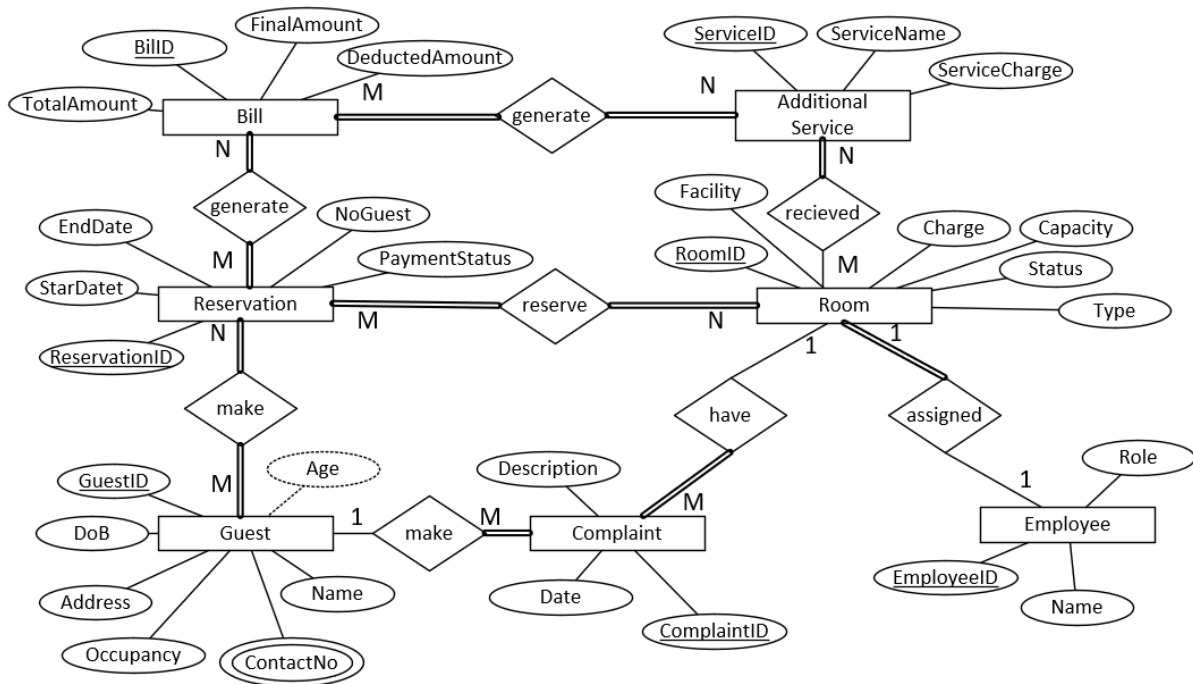


Figure 1: ER Diagram

Assumptions:

- Guests can have one or more contact number.
- A guest can have multiple reservations, but each reservation is tied to single guest.
- A guest can make multiple complaints. Each complaint is tied to a specific guest and includes a Description and Date.
- A final bill is generated at check-out, summarizing all Charges including room Charges and additional services.
- A guest can reserve multiple rooms, and each reservation is treated as a separated reservation entry.
- Employees can serve multiple rooms, and each room can have multiple assigned employees.

Task 1.2

a. User and System Requirements and tables for the Hotel Management System

User Requirements:

- Record all data about guests, bills, reservations, complaints, rooms, additional services and employees.
- Add information when needed.
- Edit the information as needed.
- Delete information when necessary.
- Organized record keeping.

System Requirements:

- A person with knowledge of databases is required to maintain the database.
- A computer with the following software and hardware specifications is required.
 - 2.6 GHz or faster 64-bit processor.
 - Minimum of 4 GB of RAM.
 - Minimum of 10 GB hard disk space.
 - Operating system should be windows 7 or above.
- SQL Server Management Studio application.

tables for the above designed ER diagram

guest table

GuestID (PK)	Name	ContactNo	Occupancy	DoB	Address	Age
G001	Roy	0754419999 0774856244	Yes	1-2-2000	12 Main St, NY 10001	24
G002	Malik	0756894421	Yes	2-5-1992	34 Oak St, LA 90001	32
G003	Ahsan	0716569826	Yes	9-5-1999	56 Pine Ln, TX 73301	25
G004	Alice	0726581568	Yes	1-6-2006	78 Birch Rd, IL 60601	18
G005	Dori	0704205128	Yes	2-6-2005	43 Willow Ln, OR 97205	19

Table 2: Guest table

room table

RoomID (PK)	Facility	Type	Charge	Capacity	Status
R001	1chair, 1bed	single	500.00	1	Vacant
R002	1chair, 1bed	single	500.00	1	Occupied
R003	2bed, 2chair	double	2000.00	2	Vacant
R004	2bed, 2chair	double	2000.00	2	Vacant
R005	AC, 5bed, 5chair	family	5000.00	5	Vacant

Table 3: Room table

Reservation table

ReservationID (PK)	GuestID (FK)	RoomID (FK)	StartDate	EndDate	NoGuest	Payment Status
RES01	G001	R001	2023-06-01	2023-06-05	1	Paid
RES02	G002	R002	2023-06-03	2023-06-07	1	Paid
RES03	G003	R003	2023-06-02	2023-06-06	2	Paid
RES04	G004	R004	2023-06-04	2023-06-08	2	Paid
RES05	G005	R005	2023-06-05	2023-06-09	4	Paid

Table 4: reservation table

Employee table

EmployeeID (PK)	RoomID (FK)	Name	Role
E001	R001	Marcus	room boy
E002	R002	Shan	room boy
E003	R003	John	room boy
E004	R004	Edna	waitress
E005	R005	Evana	waitress

Table 5: Employee table

Additional Service table

ServiceID (PK)	RoomID (FK)	ServiceName	ServiceCharge
SER001	R001	Laundry	200.00
SER002	R002	RoomService	100.00
SER003	R003	Spa	500.00
SER004	R004	Gym	150.00
SER005	R005	Breakfast	50.00

Table 6: Additional Service table

Bill table

BillID (PK)	ServiceID (FK)	ReservationID (FK)	TotalAmount	DeductedAmount	FinalAmount
B01	SER001	RES01	700.00	100.00	600.00
B02	SER002	RES02	1100.00	100.00	1000.00
B03	SER003	RES03	2500.00	100.00	2400.00
B04	SER004	RES04	3150.00	100.00	3050.00
B05	SER005	RES05	5050.00	100.00	4950.00

Table 7: Bill table

Complaint table

ComplaintID (PK)	GuestID (FK)	RoomID (FK)	Description	Date
C001	G001, G007	R001	Room not clean	2023-06-01
C002	G002	R002	No hot water	2023-06-03
C003	G003	R003	Bad food quality	2023-06-02
C004	G004	R004	No cold water	2023-06-04
C005	G005, G008	R005	AC not working	2023-06-05

Table 8: Complaint table

Data validation for the tables

For the above tables we created, we should give appropriate data Types for the values.

- We can give **char** data Type for the values of the attributes like **GuestID, RoomID, ReservationID, EmployeeID, ServiceID, BillID, ComplaintID**.
- We can give **varchar** data Type for the values like **Name of guest, Address, room Facility, room Type, room Status, payment Status, Name of employee, Role of employee, service Name and complaint Description**.
- We can give **int** data Type for the values like **contact number, room Capacity, number of guests**.
- We can give **Date** data Type for the values like **Date of birth of guest and reservation start and end Date**.
- We can give **currency** data Type for the values like **service Charge, deducted amount, total amount and final amount**.

b. Applying the Normalization rules on the above created tables up to third normal form(3NF).

Database normalization is a method for structuring data within a database to guarantee clear and accurate results. This fundamental concept in relational databases can involve duplicating information and creating extra tables. (Margaret Rouse, 2019)

1. Guest table

GuestID (PK)	Name	ContactNo	Occupancy	DoB	Address	Age
G001	Roy	0754419999 0774856244	Yes	1-2-2000	12 Main St, NY 10001	24
G002	Malik	0756894421	Yes	2-5-1992	34 Oak St, LA 90001	32
G003	Ahsan	0716569826	Yes	9-5-1999	56 Pine Ln, TX 73301	25
G004	Alice	0726581568	Yes	1-6-2006	78 Birch Rd, IL 60601	18
G005	Dori	0704205128	Yes	2-6-2005	43 Willow Ln, OR 97205	19

Table 9: Unnormalized guest table

Step 1: First Normal Form (1NF)

Rules:

- There are no repeating or duplicate fields.
- Each cell contains only a single value.

In the guest table there are multi-values. So, we have to separate them into new rows.

GuestID (PK)	Name	ContactNo	Occupancy	DoB	Address	Age
G001	Roy	0754419999	Yes	1-2-2000	12 Main St, NY 10001	24
G001	Roy	0774856244	Yes	1-2-2000	12 Main St, NY 10001	24
G002	Malik	0756894421	Yes	2-5-1992	34 Oak St, LA 90001	32
G003	Ahsan	0716569826	Yes	9-5-1999	56 Pine Ln, TX 73301	25
G004	Alice	0726581568	Yes	1-6-2006	78 Birch Rd, IL 60601	18
G005	Dori	0704205128	Yes	2-6-2005	43 Willow Ln, OR 97205	19

Table 10: 1NF guest table

Step 2: Second Normal Form (2NF)

Rules:

- All non-key fields depend on all components of the primary key.

In our latest normalized room table, not all non-key fields depend on all components of the primary key. In order to that we have to normalize this table to 2NF. To do this we have separate the table into two tables.

Guest table

GuestID (PK)	Name	Occupancy	DoB	Address	Age
G001	Roy	Yes	1-2-2000	12 Main St, NY 10001	24
G001	Roy	Yes	1-2-2000	12 Main St, NY 10001	24
G002	Malik	Yes	2-5-1992	34 Oak St, LA 90001	32
G003	Ahsan	Yes	9-5-1999	56 Pine Ln, TX 73301	25
G004	Alice	Yes	1-6-2006	78 Birch Rd, IL 60601	18
G005	Dori	Yes	2-6-2005	43 Willow Ln, OR 97205	19

Table 11: Guest table

Guest contacts table

GuestID (PK)	ContactNo (PK)
G001	0754419999
G001	0774856244
G002	0756894421
G003	0716569826
G004	0726581568
G005	0704205128

Table 12: Guest contacts table

Step 3: Third normal form (3NF)

Rules:

- No non-key field depends upon another.

In our latest normalized guest table, no non-key field depends upon each other. In order to that we don't have to normalized this table. Its already in 3NF.

2.room table

RoomID (PK)	Facility	Type	Charge	Capacity	Status
R001	1chair, 1bed	Single	500.00	1	Vacant
R002	1chair, 1bed	single	500.00	1	Occupied
R003	2bed, 2chair	double	2000.00	2	Vacant
R004	2bed, 2chair	double	2000.00	2	vacant
R005	AC, 5bed, 5chair	family	5000.00	5	Vacant

Table 13: unnormalized room table

Step 1: First Normal Form (1NF)

Rules:

- There are no repeating or duplicate fields.
- Each cell contains only a single value.

In the room table there are multi-values. We have to separate them into new rows.

RoomID (PK)	Facility	Type	Charge	Capacity	Status
R001	1chair	single	500.00	1	Vacant
R001	1bed	single	500.00	1	Vacant
R002	1chair	single	500.00	1	Occupied
R002	1bed	single	500.00	1	Occupied
R003	2bed	double	2000.00	2	Vacant
R003	2chair	double	2000.00	2	Vacant
R004	2bed	double	2000.00	2	Vacant
R004	2chair	double	2000.00	2	Vacant
R005	AC	family	5000.00	5	Vacant
R005	5bed	family	5000.00	5	Vacant
R005	5chair	family	5000.00	5	Vacant

Table 14: 1NF room table

Step 2: Second Normal Form (2NF)

Rules:

- All non-key fields depend on all components of the primary key.

In our latest normalized room table, not all non-key fields depend on all components of the primary key. In order to that we have to normalize this table to 2NF. To do this we have separate the table into two tables.

room Facility

RoomID (PK)	Facility (PK)
R001	1chair
R001	1bed
R002	1chair
R002	1bed
R003	2bed
R003	2chair
R004	2bed
R004	2chair
R005	AC
R005	5bed
R005	5chair

Table 15: 2NF room table (facility)

room

RoomID (PK)	Type	Charge	Capacity	Status
R001	single	500.00	1	vacant
R002	single	500.00	1	occupied
R003	double	2000.00	2	vacant
R004	double	2000.00	2	vacant
R005	family	5000.00	5	vacant

Table 16: 2NF room table (room)

Step 3: Third normal form (3NF)

Rules:

- No non-key field depends upon another.

In our latest normalized room table, some non-key fields are depending upon each other. So, we have to separate the room detail table into two tables. We have to normalize the table like below.

room Facility

RoomID (PK)	Facility (PK)
R001	1chair
R001	1bed
R002	1chair
R002	1bed
R003	2bed
R003	2chair
R004	2bed
R004	2chair
R005	AC
R005	5bed
R005	5chair

Table 17: 3NF room table (facility)

room

RoomID (PK)	Type	Status
R001	single	vacant
R002	single	occupied
R003	double	vacant
R004	double	vacant
R005	family	vacant

Table 18: 3NF room table (room)

room detail

Type (PK)	Charge	Capacity
single	500.00	1
double	2000.00	2
family	5000.00	5

Table 19: 3NF room table (room detail)

3.reservation table

ReservationID (PK)	GuestID (FK)	RoomID (FK)	StartDate	EndDate	NoGuest	PaymentStatus
RES01	G001	R001	2023-06-01	2023-06-05	1	Paid
RES02	G002	R002	2023-06-03	2023-06-07	1	Paid
RES03	G003	R003	2023-06-02	2023-06-06	2	Paid
RES04	G004	R004	2023-06-04	2023-06-08	2	Paid
RES05	G005	R005	2023-06-05	2023-06-09	4	Paid

Table 20: unnormalized reservation table

In this reservation table, we don't need to use any normalization method. Because, its already in the 3rd normal form.

4.employee table

EmployeeID (PK)	RoomID (FK)	Name	Role	AssignedRoom
E001	R001	Marcus	room boy	101
E002	R002	Shan	room boy	102
E003	R003	John	room boy	103
E004	R004	Edna	waitress	104
E005	R005	Evana	waitress	105

Table 21: unnormalized employee table

In this employee table, we don't need to use any normalization method. Because, its already in the 3rd normal form.

5.additional services table

ServiceID (PK)	RoomID (FK)	ServiceName	ServiceCharge
SER001	R001	Laundry	200.00
SER002	R002	RoomService	100.00
SER003	R003	Spa	500.00
SER004	R004	Gym	150.00
SER005	R005	Breakfast	50.00

Table 22: unnormalized additional service table

In this additional services table, we don't need to use any normalization method. Because, its already in the 3rd normal form.

6.bill table

BillID (PK)	ServiceID (FK)	ReservationID (FK)	TotalAmount	DeductedAmount	FinalAmount
B01	SER001	RES01	700.00	100.00	600.00
B02	SER002	RES02	1100.00	100.00	1000.00
B03	SER003	RES03	2500.00	100.00	2400.00
B04	SER004	RES04	3150.00	100.00	3050.00
B05	SER005	RES05	5050.00	100.00	4950.00

Table 23: unnormalized bill table

In this bill table we don't need to use any normalization method. Because, its already in the 3rd normal form.

7.complaint table

ComplaintID (PK)	GuestID (FK)	RoomID (FK)	Description	Date
C001	G001	R001	Room not clean	2023-06-01
C002	G002	R002	No hot water	2023-06-03
C003	G003	R003	Bad food quality	2023-06-02
C004	G004	R004	No cold water	2023-06-04
C005	G005	R005	AC not working	2023-06-05

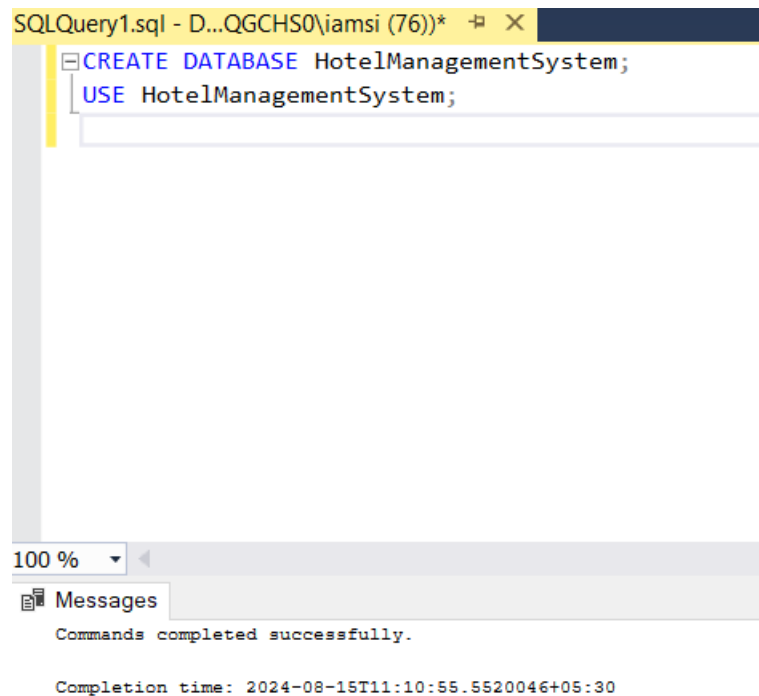
Table 24: unnormalized complaint table

In this additional services table, we don't need to use any normalization method. Because, its already in the 3rd normal form.

Task 2

2.1 Develop the physical relational database with MSSQL Server and identifying and assigning primary and foreign keys properly.

1. Create the Database



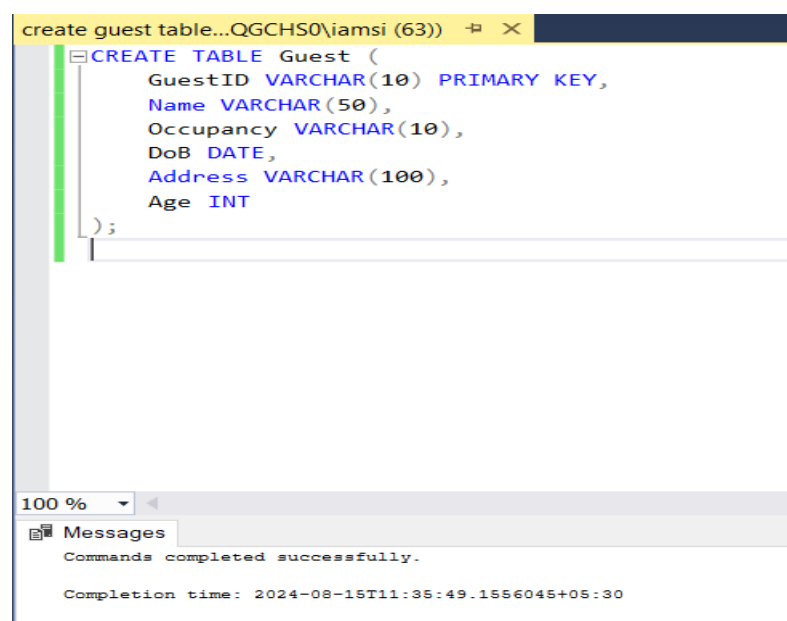
The screenshot shows a SQL query window titled 'SQLQuery1.sql - D:\...QGCHS0\iamsi (76))*'. The query contains the following commands:

```
CREATE DATABASE HotelManagementSystem;
USE HotelManagementSystem;
```

The 'Messages' pane at the bottom indicates that the commands were completed successfully. The completion time is 2024-08-15T11:10:55.5520046+05:30.

Figure 2:create database

2. Create the Guest Table



The screenshot shows a SQL query window titled 'create guest table...QGCHS0\iamsi (63))'. The query contains the following command:

```
CREATE TABLE Guest (
    GuestID VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(50),
    Occupancy VARCHAR(10),
    DoB DATE,
    Address VARCHAR(100),
    Age INT
);
```

The 'Messages' pane at the bottom indicates that the commands were completed successfully. The completion time is 2024-08-15T11:35:49.1556045+05:30.

Figure 3:create guest table

3. Create the Guest Contacts Table

```
create guest conta...QGCHS0\iamsi (53))
CREATE TABLE GuestContacts (
    GuestID VARCHAR(10),
    ContactNo VARCHAR(15),
    PRIMARY KEY (GuestID, ContactNo),
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID)
);
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-08-15T11:38:19.5889320+05:30

Figure 4:create guest contacts table

4. Create the RoomFacility Table

```
create room facili...LQGCHS0\iamsi (54))
CREATE TABLE RoomFacility (
    RoomID VARCHAR(10),
    Facility VARCHAR(50),
    PRIMARY KEY (RoomID, Facility)
);
```

100 %

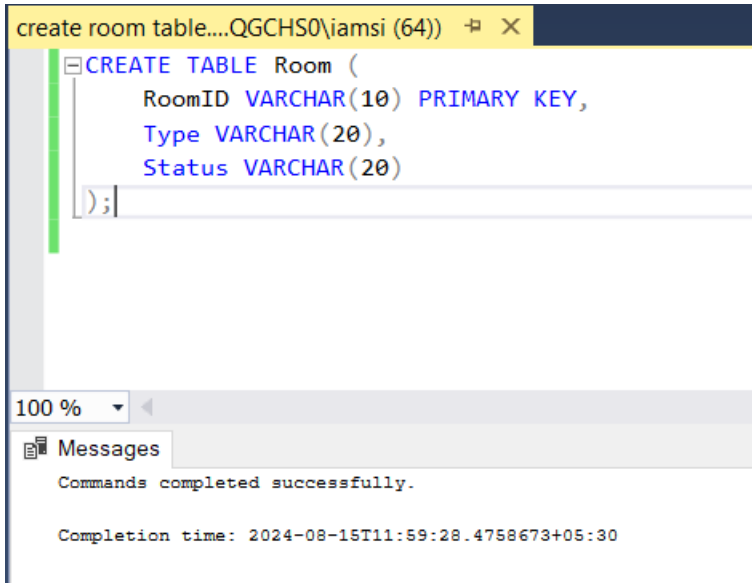
Messages

Commands completed successfully.

Completion time: 2024-08-15T11:43:26.4186322+05:30

Figure 5:create room facility table

5. Create the Room Table



```
create room table...QGCHS0\iamsi (64) X
```

```
CREATE TABLE Room (  
    RoomID VARCHAR(10) PRIMARY KEY,  
    Type VARCHAR(20),  
    Status VARCHAR(20)  
);
```

100 %

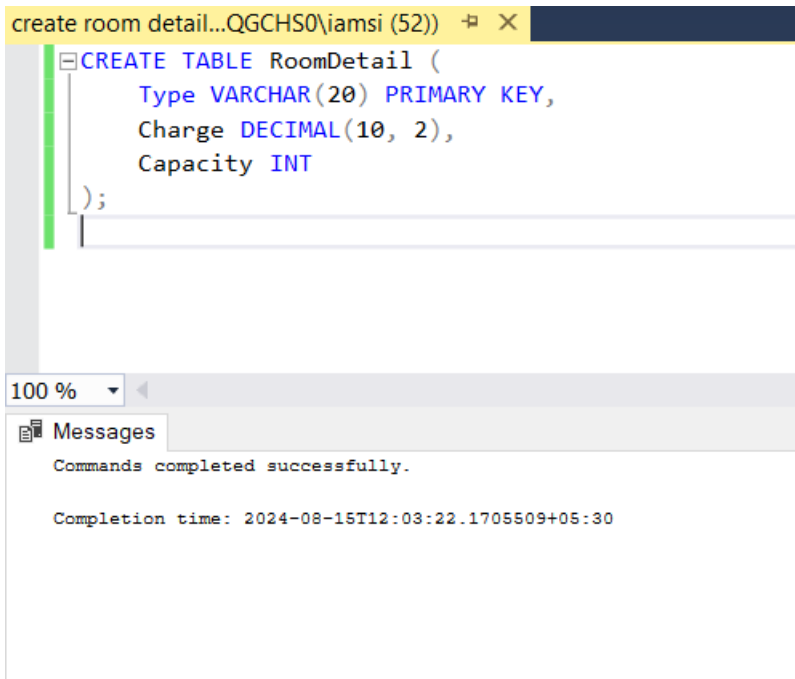
Messages

Commands completed successfully.

Completion time: 2024-08-15T11:59:28.4758673+05:30

Figure 6: create room table

6. Create the RoomDetail Table



```
create room detail...QGCHS0\iamsi (52) X
```

```
CREATE TABLE RoomDetail (  
    Type VARCHAR(20) PRIMARY KEY,  
    Charge DECIMAL(10, 2),  
    Capacity INT  
);
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-08-15T12:03:22.1705509+05:30

Figure 7: create room detail table

7. Create the Reservation Table

```
create reservation...QGCHS0\iamsi (63))  X
CREATE TABLE Reservation (
    ReservationID VARCHAR(10) PRIMARY KEY,
    GuestID VARCHAR(10),
    RoomID VARCHAR(10),
    StartDate DATE,
    EndDate DATE,
    NoGuest INT,
    PaymentStatus VARCHAR(20),
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID),
    FOREIGN KEY (RoomID) REFERENCES Room(RoomID)
);
```

100 %

 Messages

Commands completed successfully.

Completion time: 2024-08-15T12:06:41.5104017+05:30

Figure 8:create reservation table

8. Create the Employee Table

```
create employee t...QGCHS0\iamsi (51))  X
CREATE TABLE Employee (
    EmployeeID VARCHAR(10) PRIMARY KEY,
    RoomID VARCHAR(10),
    Name VARCHAR(50),
    Role VARCHAR(30),
    AssignedRoom VARCHAR(10),
    FOREIGN KEY (RoomID) REFERENCES Room(RoomID)
);
```

100 %

 Messages

Commands completed successfully.

Completion time: 2024-08-15T12:08:36.2921891+05:30

Figure 9:create employee table

9. Create the AdditionalServices Table

```
create additional...LQGCHS0\iamsi (72)) + X
CREATE TABLE AdditionalServices (
    ServiceID VARCHAR(10) PRIMARY KEY,
    RoomID VARCHAR(10),
    ServiceName VARCHAR(50),
    ServiceCharge DECIMAL(10, 2),
    FOREIGN KEY (RoomID) REFERENCES Room(RoomID)
);
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-08-15T12:09:58.9864235+05:30

Figure 10: create additional services table

10. Create the Bill Table

```
create bill table.s...LQGCHS0\iamsi (51)) + X
CREATE TABLE Bill (
    BillID VARCHAR(10) PRIMARY KEY,
    ServiceID VARCHAR(10),
    ReservationID VARCHAR(10),
    TotalAmount DECIMAL(10, 2),
    DeductedAmount DECIMAL(10, 2),
    FinalAmount DECIMAL(10, 2),
    FOREIGN KEY (ServiceID) REFERENCES AdditionalServices(ServiceID),
    FOREIGN KEY (ReservationID) REFERENCES Reservation(ReservationID)
);
```

100 %

Messages

Commands completed successfully.

Completion time: 2024-08-15T12:12:32.8034611+05:30

Figure 11: create bill table

11. Create the Complaints Table

```
create complaints...QGCHS0\iamsi (51))  + X
```

```
CREATE TABLE Complaints (
    ComplaintID VARCHAR(10) PRIMARY KEY,
    GuestID VARCHAR(10),
    RoomID VARCHAR(10),
    Description VARCHAR(255),
    Date DATE,
    FOREIGN KEY (GuestID) REFERENCES Guest(GuestID),
    FOREIGN KEY (RoomID) REFERENCES Room(RoomID)
);
```

100 %

 Messages

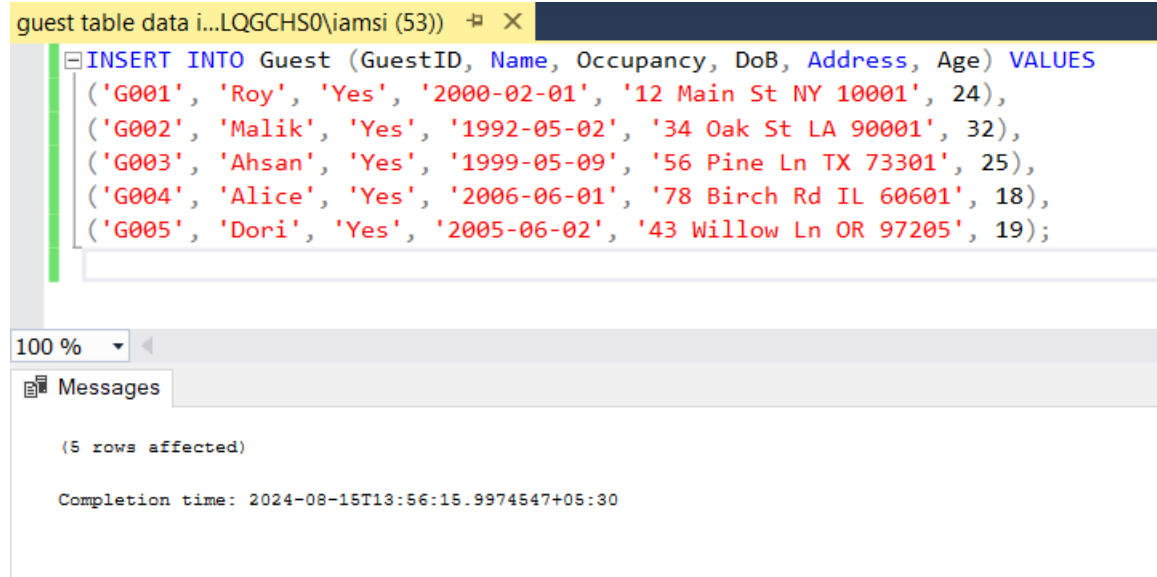
Commands completed successfully.

Completion time: 2024-08-15T16:24:53.6938145+05:30

Figure 12:create complaint table

2.2. Implement the query language to insert data to the above created tables

1. Insert Data into the Guest Table



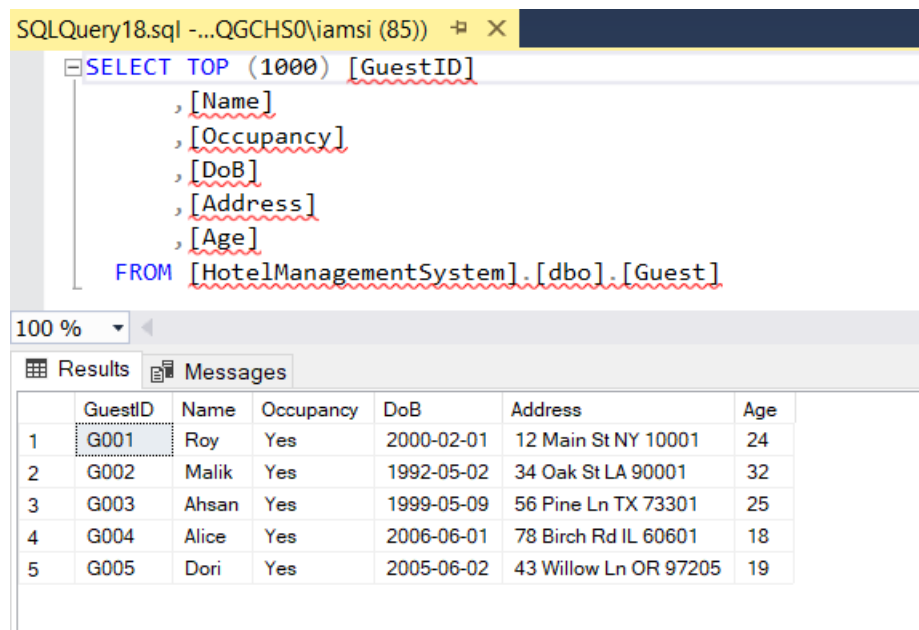
```

guest table data i...LQGCHS0\iamsi (53)
INSERT INTO Guest (GuestID, Name, Occupancy, DoB, Address, Age) VALUES
('G001', 'Roy', 'Yes', '2000-02-01', '12 Main St NY 10001', 24),
('G002', 'Malik', 'Yes', '1992-05-02', '34 Oak St LA 90001', 32),
('G003', 'Ahsan', 'Yes', '1999-05-09', '56 Pine Ln TX 73301', 25),
('G004', 'Alice', 'Yes', '2006-06-01', '78 Birch Rd IL 60601', 18),
('G005', 'Dori', 'Yes', '2005-06-02', '43 Willow Ln OR 97205', 19);

100 %
Messages
(5 rows affected)
Completion time: 2024-08-15T13:56:15.9974547+05:30

```

Figure 13:insert data into guest table



```

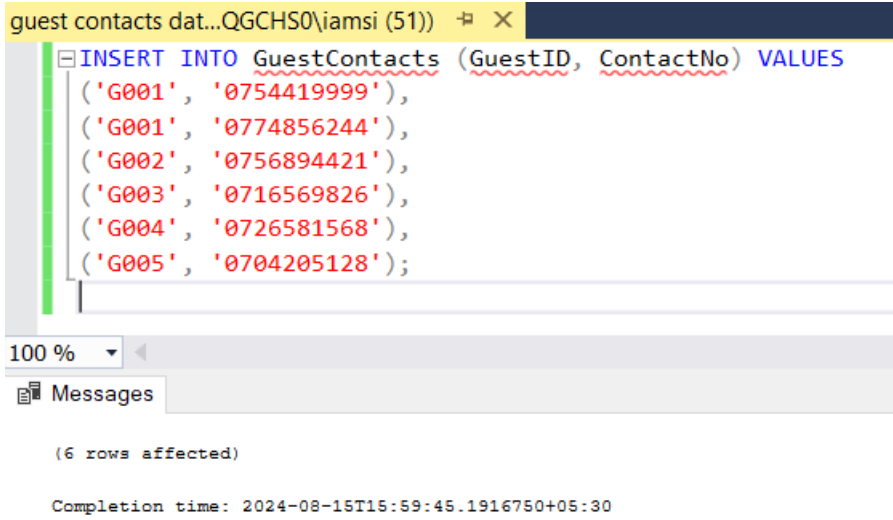
SQLQuery18.sql -...QGCHS0\iamsi (85)
SELECT TOP (1000) [GuestID]
, [Name]
, [Occupancy]
, [DoB]
, [Address]
, [Age]
FROM [HotelManagementSystem].[dbo].[Guest]

```

	GuestID	Name	Occupancy	DoB	Address	Age
1	G001	Roy	Yes	2000-02-01	12 Main St NY 10001	24
2	G002	Malik	Yes	1992-05-02	34 Oak St LA 90001	32
3	G003	Ahsan	Yes	1999-05-09	56 Pine Ln TX 73301	25
4	G004	Alice	Yes	2006-06-01	78 Birch Rd IL 60601	18
5	G005	Dori	Yes	2005-06-02	43 Willow Ln OR 97205	19

Figure 14:viewing inserted data in guest table

2. Insert Data into the GuestContacts Table



```

INSERT INTO GuestContacts (GuestID, ContactNo) VALUES
('G001', '0754419999'),
('G001', '0774856244'),
('G002', '0756894421'),
('G003', '0716569826'),
('G004', '0726581568'),
('G005', '0704205128');
  
```

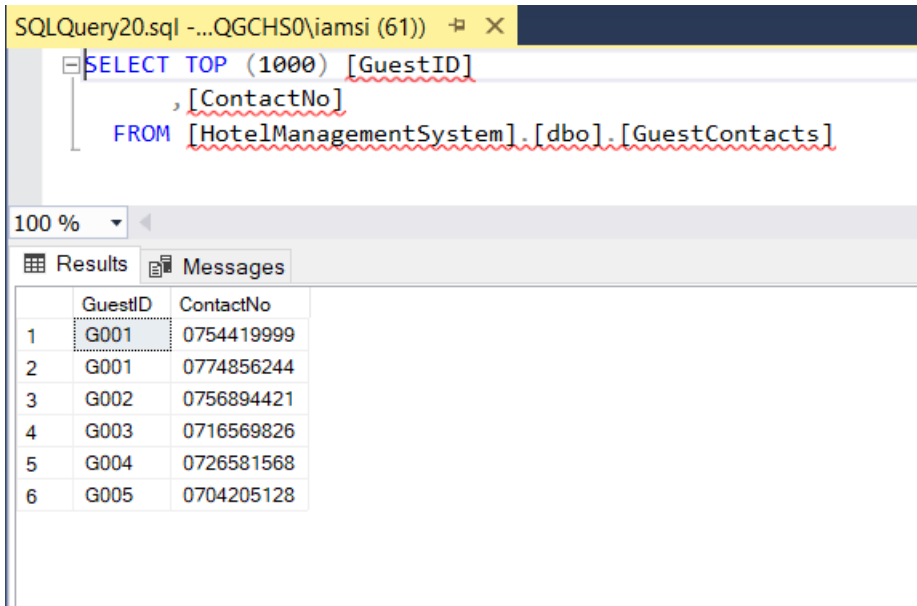
100 %

Messages

(6 rows affected)

Completion time: 2024-08-15T15:59:45.1916750+05:30

Figure 15: insert data into the guest contacts table



```

SELECT TOP (1000) [GuestID]
, [ContactNo]
FROM [HotelManagementSystem].[dbo].[GuestContacts]
  
```

100 %

Results Messages

	GuestID	ContactNo
1	G001	0754419999
2	G001	0774856244
3	G002	0756894421
4	G003	0716569826
5	G004	0726581568
6	G005	0704205128

Figure 16: viewing inserted data into the guest contacts table

3. Insert Data into the RoomFacility Table

room facility data...QGCHS0\iamsi (51) + X

```

INSERT INTO RoomFacility (RoomID, Facility) VALUES
('R001', '1chair'),
('R001', '1bed'),
('R002', '1chair'),
('R002', '1bed'),
('R003', '2bed'),
('R003', '2chair'),
('R004', '2bed'),
('R004', '2chair'),
('R005', 'AC'),
('R005', '5bed'),
('R005', '5chair');

```

100 %

Messages

(11 rows affected)

Completion time: 2024-08-15T16:01:30.2328295+05:30

Figure 17:insert data into room facility table

SQLQuery22.sql -...QGCHS0\iamsi (61) + X

```

SELECT TOP (1000) [RoomID]
, [Facility]
FROM [HotelManagementSystem].[dbo].[RoomFacility]

```

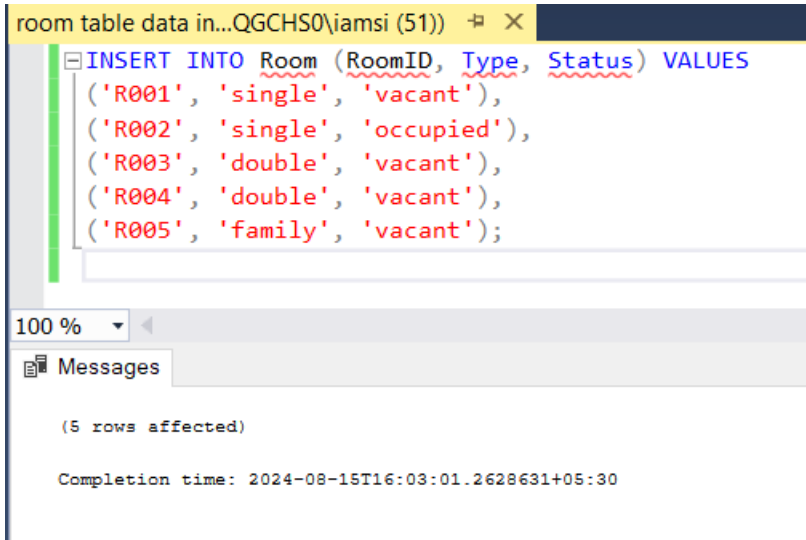
100 %

Results Messages

	RoomID	Facility
1	R001	1bed
2	R001	1chair
3	R002	1bed
4	R002	1chair
5	R003	2bed
6	R003	2chair
7	R004	2bed
8	R004	2chair
9	R005	5bed
10	R005	5chair
11	R005	AC

Figure 18: viewing inserted data into the room facility table

4. Insert Data into the Room Table

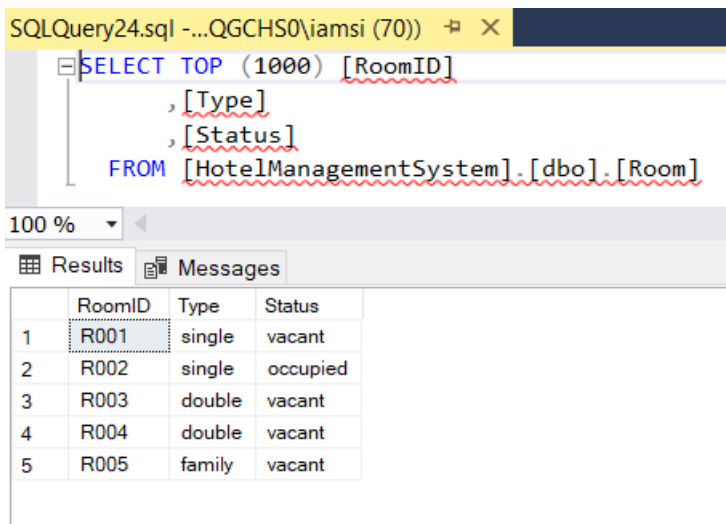


The screenshot shows a SQL query window titled "room table data in...QGCHS0\iamsi (51)". The query is an INSERT INTO statement for the Room table, inserting five rows of data. The columns are RoomID, Type, and Status. The data being inserted is as follows:

RoomID	Type	Status
R001	single	vacant
R002	single	occupied
R003	double	vacant
R004	double	vacant
R005	family	vacant

Below the query window, the Messages pane shows "(5 rows affected)" and the Completion time: 2024-08-15T16:03:01.2628631+05:30.

Figure 19:insert data into the room table

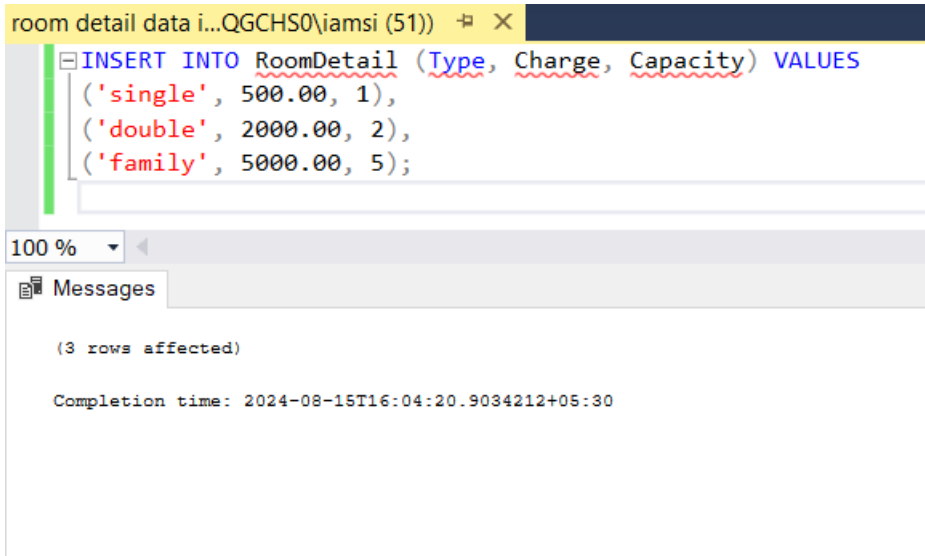


The screenshot shows a SQL query window titled "SQLQuery24.sql -...QGCHS0\iamsi (70)". The query is a SELECT TOP (1000) statement for the Room table, selecting the columns RoomID, Type, and Status. The data is retrieved from the HotelManagementSystem.dbo.Room table. The Results pane shows the following data:

	RoomID	Type	Status
1	R001	single	vacant
2	R002	single	occupied
3	R003	double	vacant
4	R004	double	vacant
5	R005	family	vacant

Figure 20:viewing inserted data into the room table

5. Insert Data into the RoomDetail Table



The screenshot shows a SQL query window with the following text:

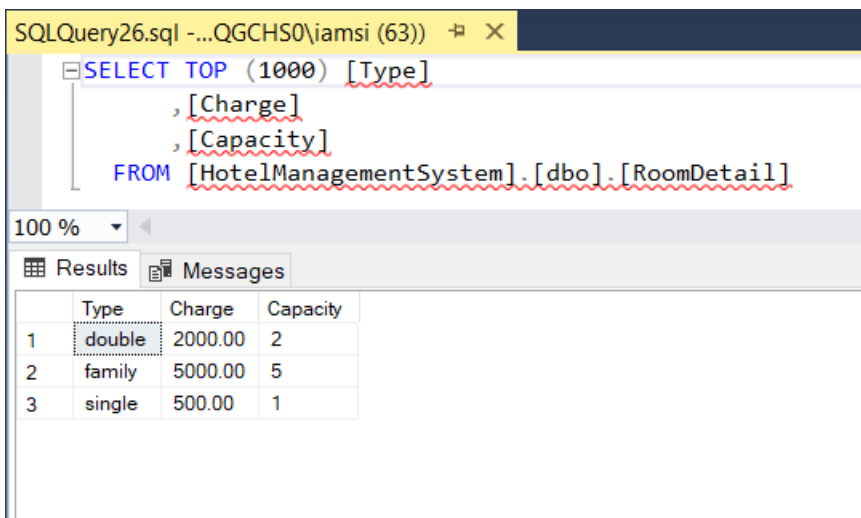
```
room detail data i...QGCHS0\iamsi (51))
INSERT INTO RoomDetail (Type, Charge, Capacity) VALUES
('single', 500.00, 1),
('double', 2000.00, 2),
('family', 5000.00, 5);
```

Below the query, the Messages pane shows:

```
(3 rows affected)

Completion time: 2024-08-15T16:04:20.9034212+05:30
```

Figure 21:insert data into the room detail table



The screenshot shows a SQL query window with the following text:

```
SQLQuery26.sql -...QGCHS0\iamsi (63))
SELECT TOP (1000) [Type]
, [Charge]
, [Capacity]
FROM [HotelManagementSystem].[dbo].[RoomDetail]
```

Below the query, the Results pane shows the following data:

	Type	Charge	Capacity
1	double	2000.00	2
2	family	5000.00	5
3	single	500.00	1

Figure 22:viewing data into the room detail table

6. Insert Data into the Reservation Table

```
reservation table...LQGCHS0\iamsi (51) X
INSERT INTO Reservation (ReservationID, GuestID, RoomID, StartDate, EndDate, NoGuest, PaymentStatus) VALUES
('RES01', 'G001', 'R001', '2023-06-01', '2023-06-05', 1, 'Paid'),
('RES02', 'G002', 'R002', '2023-06-03', '2023-06-07', 1, 'Paid'),
('RES03', 'G003', 'R003', '2023-06-02', '2023-06-06', 2, 'Paid'),
('RES04', 'G004', 'R004', '2023-06-04', '2023-06-08', 2, 'Paid'),
('RES05', 'G005', 'R005', '2023-06-05', '2023-06-09', 4, 'Paid');

100 %
Messages
(5 rows affected)
Completion time: 2024-08-15T16:05:54.9113343+05:30
```

Figure 23:insert data into the reservation table

SQLQuery28.sql -...QGCHS0\iamsi (76) X

```
SELECT TOP (1000) [ReservationID]
, [GuestID]
, [RoomID]
, [StartDate]
, [EndDate]
, [NoGuest]
, [PaymentStatus]
FROM [HotelManagementSystem].[dbo].[Reservation]
```

100 %

Results Messages

	ReservationID	GuestID	RoomID	StartDate	EndDate	NoGuest	PaymentStatus
1	RES01	G001	R001	2023-06-01	2023-06-05	1	Paid
2	RES02	G002	R002	2023-06-03	2023-06-07	1	Paid
3	RES03	G003	R003	2023-06-02	2023-06-06	2	Paid
4	RES04	G004	R004	2023-06-04	2023-06-08	2	Paid
5	RES05	G005	R005	2023-06-05	2023-06-09	4	Paid

Figure 24:viewing data into the reservation table

7. Insert Data into the Employee Table

employee table da...GCHS0\iamsi (51) ✕

```

INSERT INTO Employee (EmployeeID, RoomID, Name, Role, AssignedRoom) VALUES
('E001', 'R001', 'Marcus', 'room boy', '101'),
('E002', 'R002', 'Shan', 'room boy', '102'),
('E003', 'R003', 'John', 'room boy', '103'),
('E004', 'R004', 'Edna', 'waitress', '104'),
('E005', 'R005', 'Evana', 'waitress', '105');

```

100 %

Messages

(5 rows affected)

Completion time: 2024-08-15T16:07:23.4115291+05:30

Figure 25: inserting data into the employee table

SQLQuery30.sql -...QGCHS0\iamsi (77) ✕

```

SELECT TOP (1000) [EmployeeID]
, [RoomID]
, [Name]
, [Role]
, [AssignedRoom]
FROM [HotelManagementSystem].[dbo].[Employee]

```

100 %

Results Messages

	EmployeeID	RoomID	Name	Role	AssignedRoom
1	E001	R001	Marcus	room boy	101
2	E002	R002	Shan	room boy	102
3	E003	R003	John	room boy	103
4	E004	R004	Edna	waitress	104
5	E005	R005	Evana	waitress	105

Figure 26: viewing data into the employee table

8. Insert Data into the AdditionalServices Table

additional service...LQGCHS0\iamsi (51) ✕

```

INSERT INTO AdditionalServices (ServiceID, RoomID, ServiceName, ServiceCharge) VALUES
('SER001', 'R001', 'Laundry', 200.00),
('SER002', 'R002', 'RoomService', 100.00),
('SER003', 'R003', 'Spa', 500.00),
('SER004', 'R004', 'Gym', 150.00),
('SER005', 'R005', 'Breakfast', 50.00);

```

100 %

Messages

(5 rows affected)

Completion time: 2024-08-15T16:08:48.1115794+05:30

Figure 27: insert data into the additional services table

SQLQuery32.sql -...LQGCHS0\iamsi (55) ✕

```

SELECT TOP (1000) [ServiceID]
, [RoomID]
, [ServiceName]
, [ServiceCharge]
FROM [HotelManagementSystem].[dbo].[AdditionalServices]

```

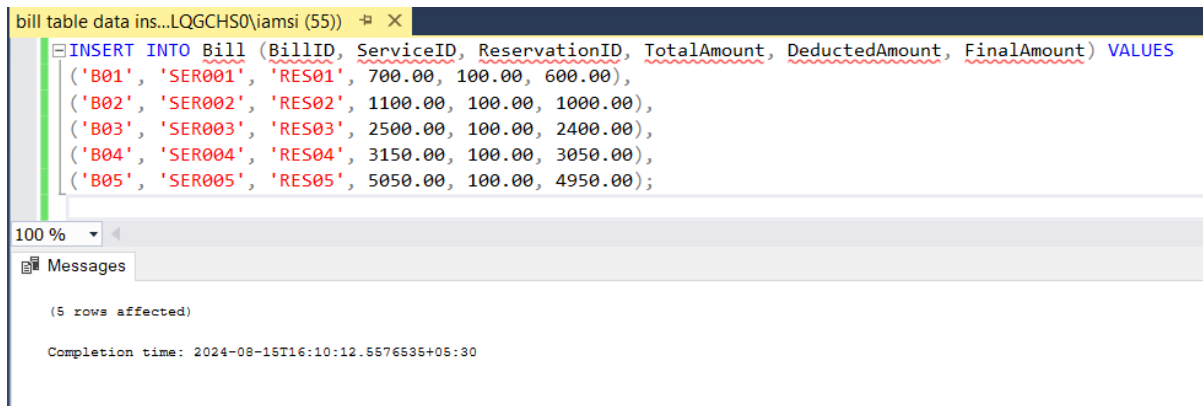
100 %

Results Messages

	ServiceID	RoomID	ServiceName	ServiceCharge
1	SER001	R001	Laundry	200.00
2	SER002	R002	RoomService	100.00
3	SER003	R003	Spa	500.00
4	SER004	R004	Gym	150.00
5	SER005	R005	Breakfast	50.00

Figure 28: viewing data into the additional services table

9. Insert Data into the Bill Table



The screenshot shows a SQL query window with the following text:

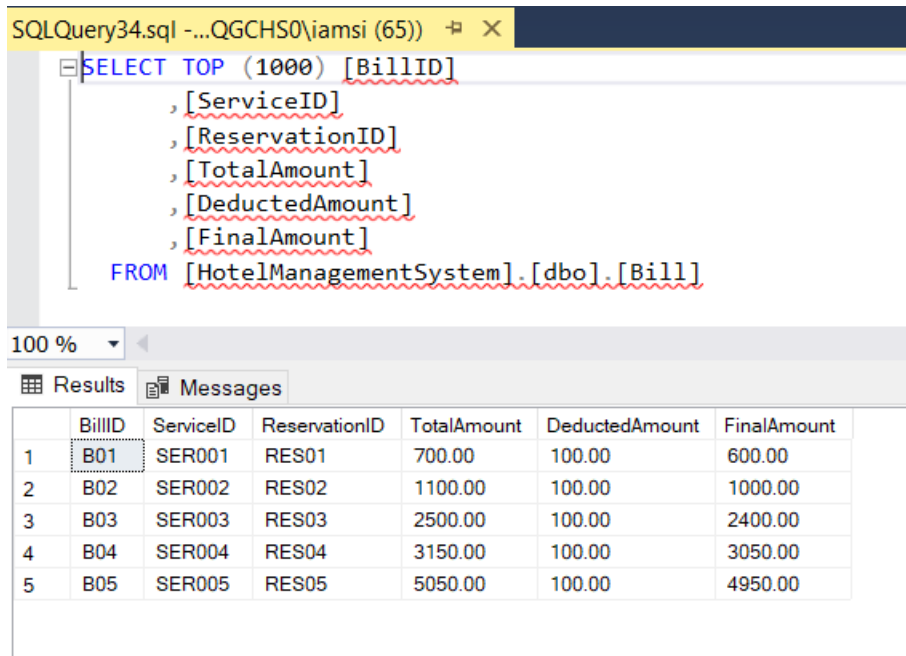
```
bill table data ins...LQGCHS0\iamsi (55)
INSERT INTO Bill (BillID, ServiceID, ReservationID, TotalAmount, DeductedAmount, FinalAmount) VALUES
('B01', 'SER001', 'RES01', 700.00, 100.00, 600.00),
('B02', 'SER002', 'RES02', 1100.00, 100.00, 1000.00),
('B03', 'SER003', 'RES03', 2500.00, 100.00, 2400.00),
('B04', 'SER004', 'RES04', 3150.00, 100.00, 3050.00),
('B05', 'SER005', 'RES05', 5050.00, 100.00, 4950.00);
```

Below the query window, the Messages pane shows:

(5 rows affected)

Completion time: 2024-08-16T16:10:12.5576535+05:30

Figure 29: insert data into the bill table



The screenshot shows a SQL query window with the following text:

```
SQLQuery34.sql -...QGCHS0\iamsi (65)
SELECT TOP (1000) [BillID]
, [ServiceID]
, [ReservationID]
, [TotalAmount]
, [DeductedAmount]
, [FinalAmount]
FROM [HotelManagementSystem].[dbo].[Bill]
```

Below the query window, the Results pane shows the following data:

	BillID	ServiceID	ReservationID	TotalAmount	DeductedAmount	FinalAmount
1	B01	SER001	RES01	700.00	100.00	600.00
2	B02	SER002	RES02	1100.00	100.00	1000.00
3	B03	SER003	RES03	2500.00	100.00	2400.00
4	B04	SER004	RES04	3150.00	100.00	3050.00
5	B05	SER005	RES05	5050.00	100.00	4950.00

Figure 30: viewing inserted data into the bill table

10. Insert Data into the Complaints Table

complaint table d...QGCHS0\iamsi (51))

```

INSERT INTO Complaints (ComplaintID, GuestID, RoomID, Description, Date) VALUES
('C001', 'G001', 'R001', 'Room not clean', '2023-06-01'),
('C002', 'G002', 'R002', 'No hot water', '2023-06-03'),
('C003', 'G003', 'R003', 'Bad food quality', '2023-06-02'),
('C004', 'G004', 'R004', 'No cold water', '2023-06-04'),
('C005', 'G005', 'R005', 'AC not working', '2023-06-05');
  
```

100 %

Messages

(5 rows affected)

Completion time: 2024-08-15T16:26:42.2371126+05:30

Figure 31:insert data into the complaint table

SQLQuery41.sql -...QGCHS0\iamsi (85))

```

SELECT TOP (1000) [ComplaintID]
, [GuestID]
, [RoomID]
, [Description]
, [Date]
FROM [HotelManagementSystem].[dbo].[Complaints]
  
```

100 %

Results Messages

	ComplaintID	GuestID	RoomID	Description	Date
1	C001	G001	R001	Room not clean	2023-06-01
2	C002	G002	R002	No hot water	2023-06-03
3	C003	G003	R003	Bad food quality	2023-06-02
4	C004	G004	R004	No cold water	2023-06-04
5	C005	G005	R005	AC not working	2023-06-05

Figure 32:viewing inserted data into the complaint table

apply the queries to update and delete.

1. Update a Guest's Address

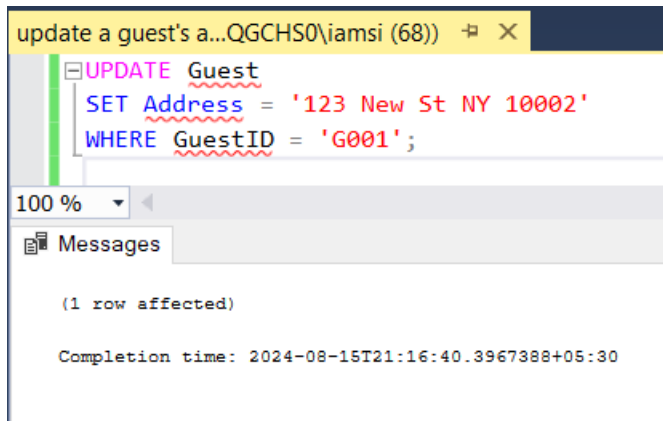


Figure 33:update a guest's address

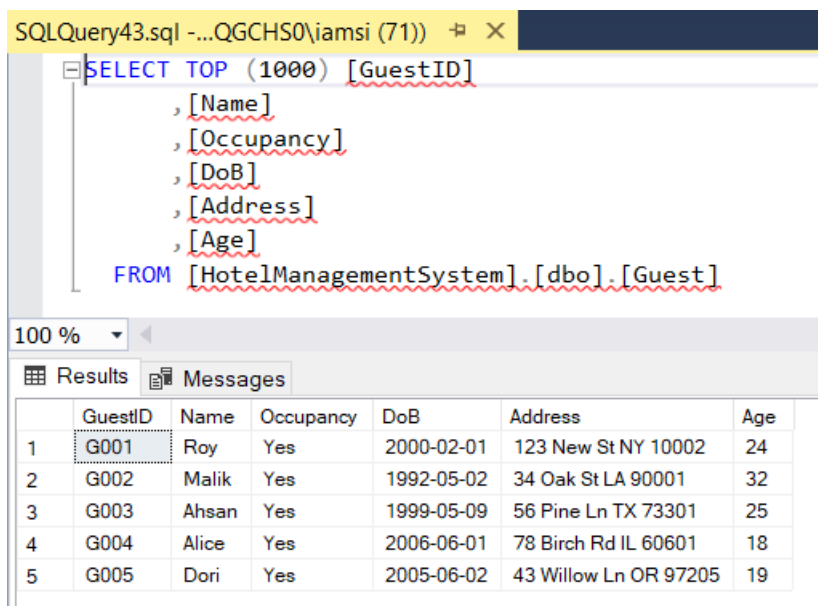
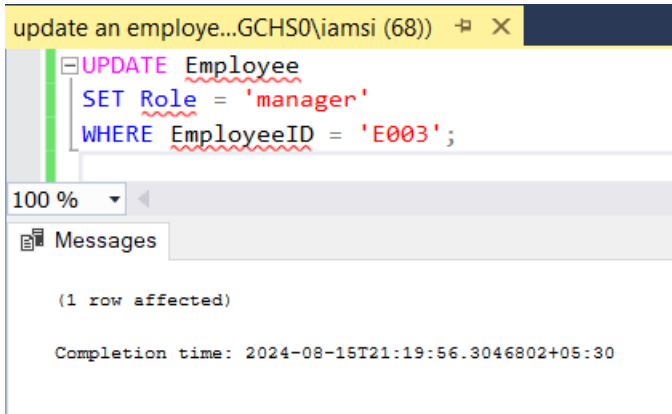


Figure 34:after update the guest's address

2. Update an Employee's Role



```

update an employee...GCHS0\iamsi (68))
UPDATE Employee
SET Role = 'manager'
WHERE EmployeeID = 'E003';
  
```

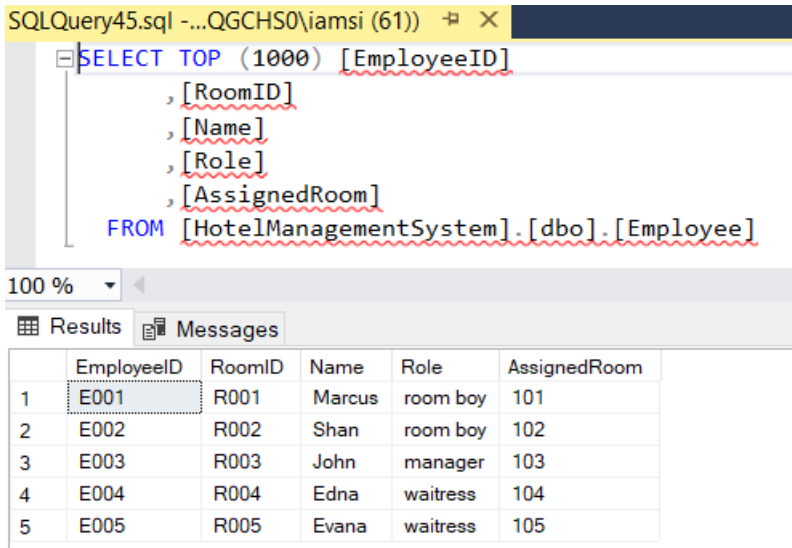
100 %

Messages

(1 row affected)

Completion time: 2024-08-15T21:19:56.3046802+05:30

Figure 35:update an employee's role



```

SQLQuery45.sql -...QGCHS0\iamsi (61))
SELECT TOP (1000) [EmployeeID]
, [RoomID]
, [Name]
, [Role]
, [AssignedRoom]
FROM [HotelManagementSystem].[dbo].[Employee]
  
```

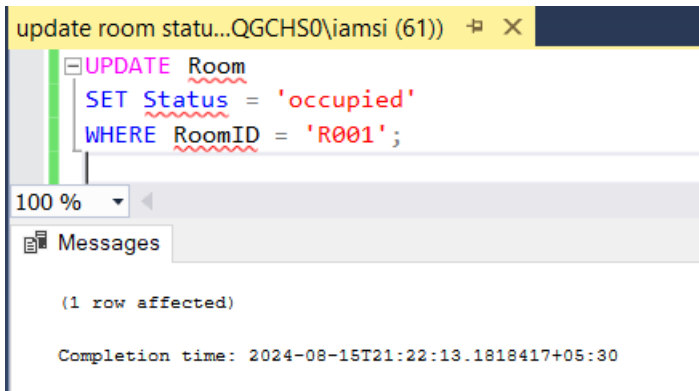
100 %

Results Messages

	EmployeeID	RoomID	Name	Role	AssignedRoom
1	E001	R001	Marcus	room boy	101
2	E002	R002	Shan	room boy	102
3	E003	R003	John	manager	103
4	E004	R004	Edna	waitress	104
5	E005	R005	Evana	waitress	105

Figure 36:after update the employee's role

3. Update Room Status



```
update room statu...QGCHS0\iamsi (61))
UPDATE Room
SET Status = 'occupied'
WHERE RoomID = 'R001';
```

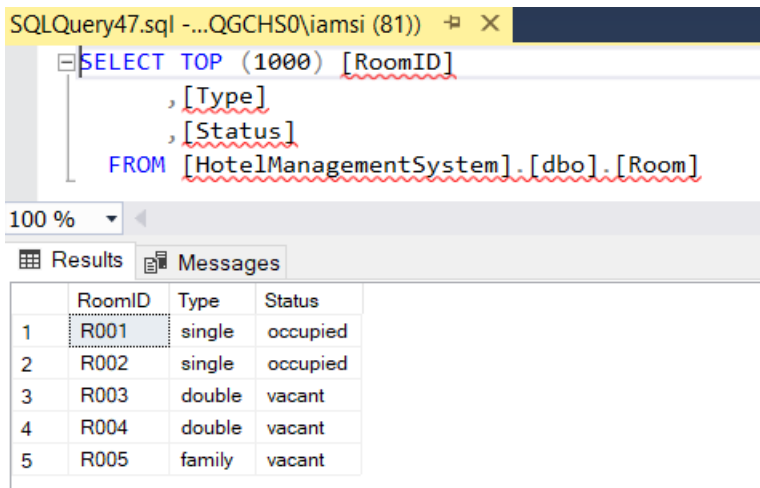
100 %

Messages

(1 row affected)

Completion time: 2024-08-15T21:22:13.1818417+05:30

Figure 37:update room status



```
SQLQuery47.sql -...QGCHS0\iamsi (81))
SELECT TOP (1000) [RoomID]
, [Type]
, [Status]
FROM [HotelManagementSystem].[dbo].[Room]
```

100 %

Results Messages

	RoomID	Type	Status
1	R001	single	occupied
2	R002	single	occupied
3	R003	double	vacant
4	R004	double	vacant
5	R005	family	vacant

Figure 38:after update the room status

4. Delete a Room Facility

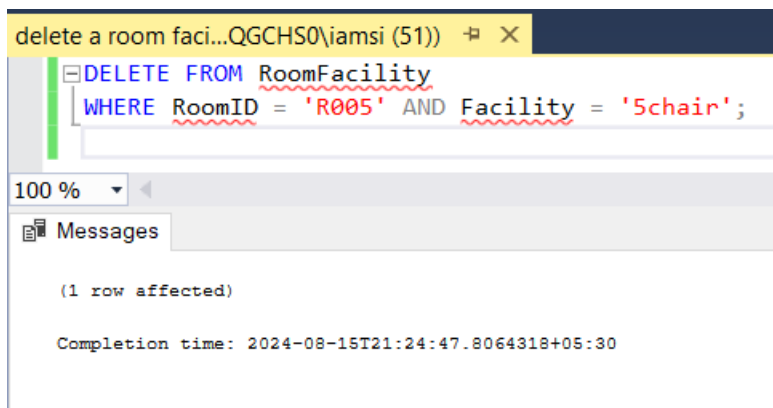


Figure 39:delete a room facility

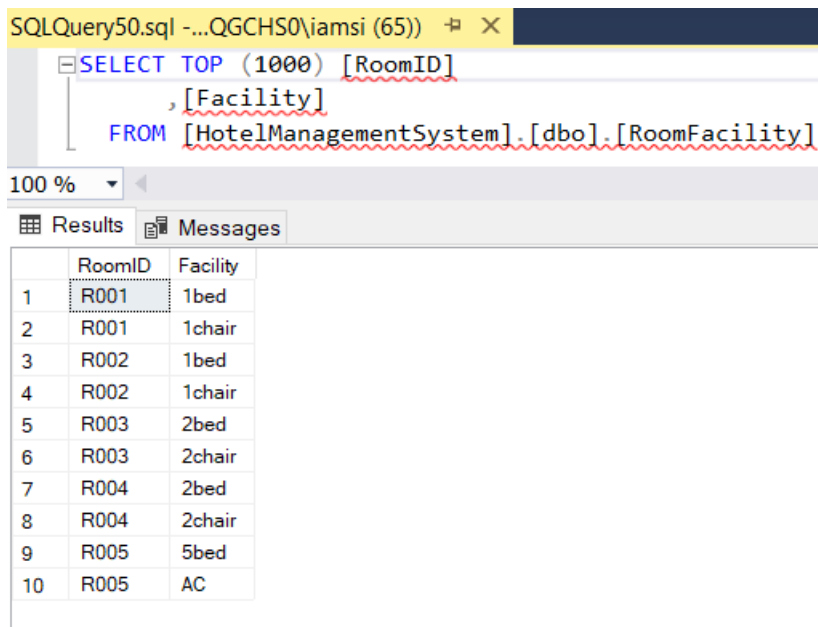


Figure 40:after delete the room facility

5. delete a complaint

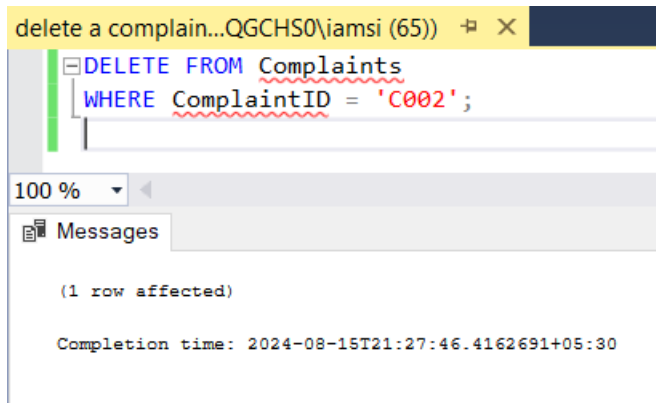


Figure 41:delete a complaint

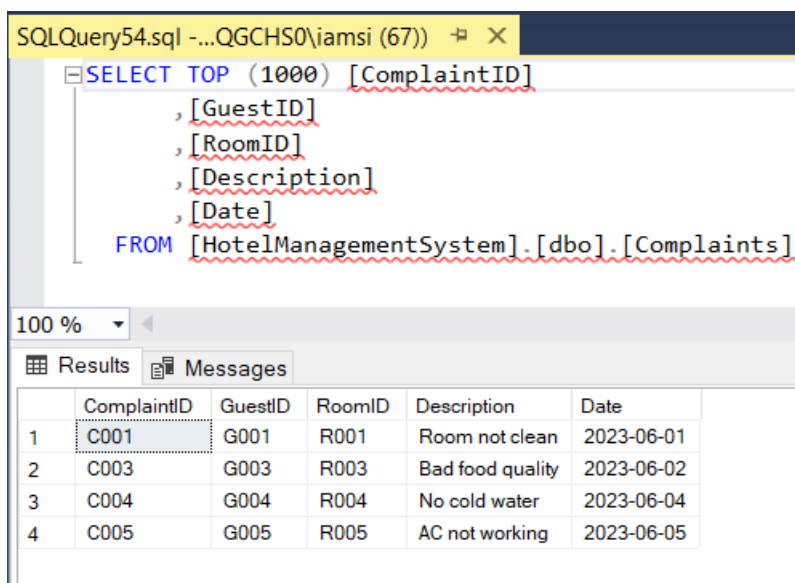
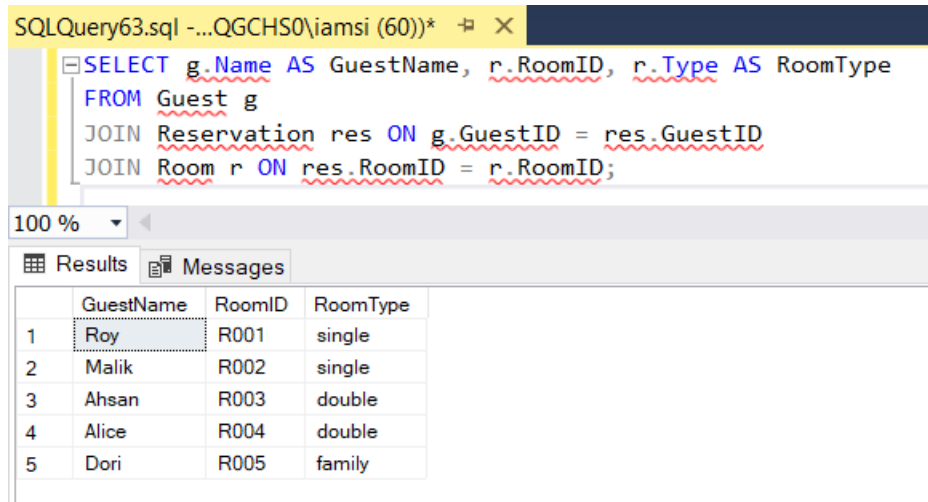


Figure 42:after delete the complaint

2.3. retrieve meaningful data using different types of complex queries.

1. JOIN

Retrieve a list of guests along with the rooms they have reserved.



The screenshot shows a SQL Query Editor window with the following query:

```
SELECT g.Name AS GuestName, r.RoomID, r.Type AS RoomType
FROM Guest g
JOIN Reservation res ON g.GuestID = res.GuestID
JOIN Room r ON res.RoomID = r.RoomID;
```

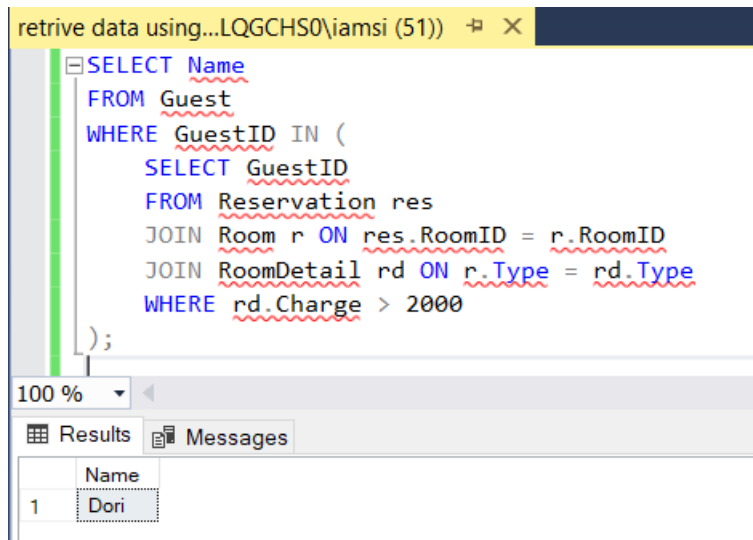
The results are displayed in a table with the following data:

	GuestName	RoomID	RoomType
1	Roy	R001	single
2	Malik	R002	single
3	Ahsan	R003	double
4	Alice	R004	double
5	Dori	R005	family

Figure 43:using join query

2. Subquery

Find all guests who have made a reservation in a room that costs more than \$2000.



The screenshot shows a SQL Query Editor window with the following query:

```
SELECT Name
FROM Guest
WHERE GuestID IN (
    SELECT GuestID
    FROM Reservation res
    JOIN Room r ON res.RoomID = r.RoomID
    JOIN RoomDetail rd ON r.Type = rd.Type
    WHERE rd.Charge > 2000
);
```

The results are displayed in a table with the following data:

	Name
1	Dori

Figure 44:using subquery

3. Aggregation

Calculate the total number of complaints per guest.

retrieve data usin...LQGCHS0\iamsi (71))

```
SELECT GuestID, COUNT(*) AS TotalComplaints
FROM Complaints
GROUP BY GuestID;
```

100 %

Results Messages

	GuestID	TotalComplaints
1	G001	1
2	G003	1
3	G004	1
4	G005	1

Figure 45:using aggregation query

4. Set Operator (UNION)

Retrieve all unique room IDs either from the RoomFacility table or the Employee table.

retrieve data usin...LQGCHS0\iamsi (51))

```
SELECT RoomID FROM RoomFacility
UNION
SELECT RoomID FROM Employee;
```

100 %

Results Messages

	RoomID
1	R001
2	R002
3	R003
4	R004
5	R005

Figure 46:using set operator

5. Unspecified WHERE Clause

Retrieve all reservations where the payment status is not specified (assuming NULL or empty string indicates unspecified).

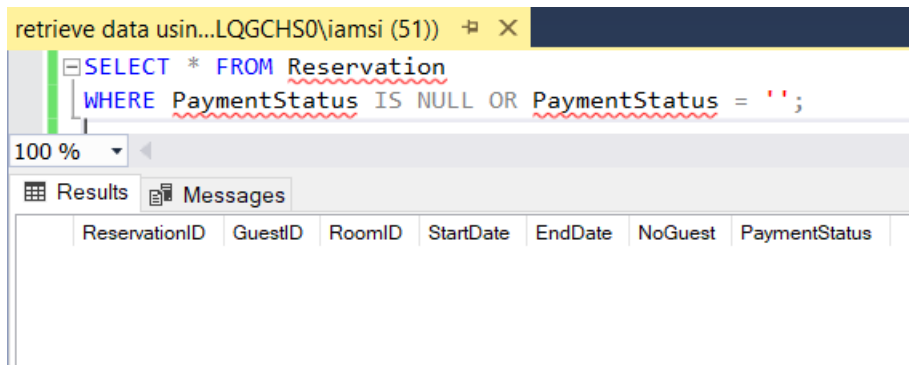


Figure 47:using unspecified where clause

6. Substring Comparison

Find all guests whose names start with the letter 'A'.

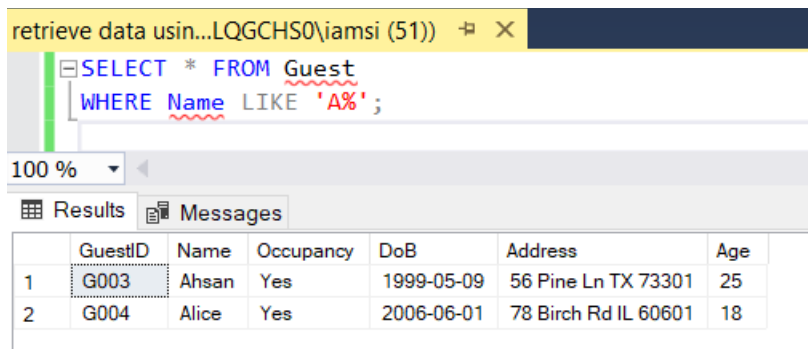


Figure 48:using substring comparison

7. Arithmetic Operator

Calculate the final amount after a 10% discount on the total amount in the Bill table

retrieve data usin...LQGCHS0\iamsi (51)) ✕

```
SELECT BillID, TotalAmount, TotalAmount * 0.9 AS DiscountedAmount
FROM Bill;
```

100 %

Results Messages

	BillID	TotalAmount	DiscountedAmount
1	B01	700.00	630.000
2	B02	1100.00	990.000
3	B03	2500.00	2250.000
4	B04	3150.00	2835.000
5	B05	5050.00	4545.000

Figure 49:using arithmetic operator

8. Wildcard

Select all columns from the Room table where the room status is 'vacant'.

retrieve data usin...LQGCHS0\iamsi (56)) ✕

```
SELECT *
FROM Room
WHERE Status = 'vacant';
```

100 %

Results Messages

	RoomID	Type	Status
1	R003	double	vacant
2	R004	double	vacant
3	R005	family	vacant

Figure 50:using wild card

Task 3

test the developed database system against user and system requirements.

Database testing

Database testing is a process that verifies the structure and content of a database. It involves examining components like schemas, tables, and triggers to ensure data accuracy and reliability. Additionally, it tests the database's performance under heavy loads by running complex queries. (Hamilton, 2024)

Types of Database testing

1. Whitebox database testing

White box testing is a method of examining the internal workings of a database. It involves testing the database's underlying structure, including tables, data models, and how different parts interact. This type of testing requires in-depth knowledge of the database and its code.

Testers look at elements like database triggers, views, and SQL queries to ensure they function correctly. They check for errors in the database's logic and structure. While it's effective at finding internal problems, it doesn't directly test how well the database responds to user actions.

2. Blackbox database testing

Black box testing focuses on checking how a database works from a user's perspective without looking at its internal code. Testers examine how data enters and leaves the database, ensuring it matches expected results. Various techniques are used to design test cases, such as considering different input values and their potential outcomes. This type of testing is done early in development and is generally less expensive than other testing methods. However, it may miss certain types of errors and doesn't provide complete coverage of the database's functionality. (Chand, 2023)

Database design and development

There are many types of Blackbox testing. But, we are going to use these 2 types for testing:

- 1. Equivalence class Partitioning:** Divides the input data into equivalent partitions, with each partition being regarded the same by the program. Testing one representative from each partition is usually enough to cover all potential scenarios.
- 2. Boundary Value Analysis:** Tests the bounds of input ranges, as errors frequently arise on the edge of input limits.

Test Plan against developed database system

Objective:

To validate the functionality of the hotel management system's database, ensuring it meets the specified requirements for data recording, updating, deleting, and retrieving across various entities.

Test Methodologies:

- **Equivalence Class Partitioning (ECP):** Identify valid and invalid partitions of input data to ensure broad coverage.
- **Boundary Value Analysis (BVA):** Focus on edge values to validate system behavior at boundaries.

Main goals of our testing

- Guest details should be saved without any errors.
- Reservations should be made without any errors.
- Checking room status feature should be work properly.
- Bills should be generated without any errors.
- Complaints should be collected properly.
- Additional services should be save properly.

1. Equivalence Class Partitioning (ECP) Tests

A. ECP Based on Valid and Invalid Input Ranges

Guest Table

Description: This test validates input fields in the Guest table by testing both valid and invalid partitions. Age should be above 18 and name should be given.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
ECP-G-001	Valid guest record	Name: Jane Doe, Age: 30, ContactNo: [555-9876]	Guest added successfully	ECP (Valid Input)	Passed
ECP-G-002	Invalid guest record with empty Name	Name: "", Age: 30, ContactNo: [555-9876]	Error: Name is required	ECP (Invalid Input)	Passed
ECP-G-003	Invalid age input	Name: Jane Doe, Age: -5, ContactNo: [555-9876]	Error: Invalid Age	ECP (Invalid Input)	Passed

Table 25: valid and invalid input test guest (ECP)

Reservation Table

Description: This test validates input fields in the Reservation table by testing both valid and invalid partitions.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
ECP-R-001	Valid reservation	StartDate: 2024-08-16, EndDate: 2024-08-20	Reservation created successfully	ECP (Valid Input)	Passed
ECP-R-002	Invalid reservation with incorrect dates	StartDate: 2024-08-20, EndDate: 2024-08-16	Error: End date must be after start date	ECP (Invalid Input)	Passed

Table 26: valid and invalid input test reservation (ECP)

Employee Table

Description: This table tracks employee details, including their roles and assigned rooms.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
ECP-E-001	Valid employee record	Name: John Smith, Role: Manager, AssignedRoom: 101	Employee added successfully	ECP (Valid Input)	Passed
ECP-E-002	Invalid employee record with missing name	Name: "", Role: Manager, AssignedRoom: 101	Error: Employee name is required	ECP (Invalid Input)	Passed
ECP-E-003	Invalid role input	Name: John Smith, Role: "", AssignedRoom: 101	Error: Role is required	ECP (Invalid Input)	Passed

Table 27:ECP based on valid and invalid input ranges in employee

B. ECP Based on Boolean Values

Room Table

Description: This test focuses on Boolean values, such as checking room availability.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
ECP-RM-001	Check room availability (Available)	RoomID: R001, Status: Available	Room is available	ECP (Valid Input)	Passed
ECP-RM-002	Check room availability (Unavailable)	RoomID: R001, Status: Unavailable	Room is unavailable	ECP (Valid Input)	Passed

Table 28:ECP based on Boolean in room table

Additional Service Table

Description: This test focuses on Boolean values related to the availability of additional services.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
ECP-AS-004	Service available (Boolean)	ServiceName: Spa, Available: True	Service is available	ECP (Valid Input)	Passed
ECP-AS-005	Service unavailable (Boolean)	ServiceName: Spa, Available: False	Service is unavailable	ECP (Valid Input)	Passed

Table 29:testing additional services ECP based on Boolean values

2. Boundary Value Analysis (BVA) Tests

A. BVA Based on Range of Values with Two Input Variables

Bill Table

Description: This test checks boundary values for the TotalAmount and DeductedAmount fields.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
BVA-B-001	Test total amount at minimum	TotalAmount: 0, DeductedAmount: 0	Bill generated successfully	BVA (Lower Boundary)	Passed
BVA-B-002	Test total amount just below minimum	TotalAmount: -1, DeductedAmount: 0	Error: Invalid total amount	BVA (Lower Boundary)	Passed
BVA-B-003	Test deducted amount just above total	TotalAmount: 500, DeductedAmount: 501	Error: Deducted amount cannot be greater than total amount	BVA (Upper Boundary)	Passed

Table 30:BVA based on range of values in billing

B. BVA of Equivalence Classes with One Input Variable

Guest Table

Description: This test examines boundary values within equivalence classes for the Age field.

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
BVA-G-001	Test age just below minimum	Age: 17	Error: Age must be ≥ 18	BVA (Lower Boundary)	Passed
BVA-G-002	Test age at minimum	Age: 18	Guest added successfully	BVA (Lower Boundary)	Passed
BVA-G-003	Test age at upper limit	Age: 120	Guest added successfully	BVA (Upper Boundary)	Passed
BVA-G-004	Test age above upper limit	Age: 121	Error: Invalid Age	BVA (Upper Boundary)	Passed

Table 31: BVA based on equivalence classes with one input in guest

C. Boundary Value Analysis (BVA) Based on Boolean Values

Complaint Table

Description: This test validates the boundary behaviour of Boolean fields (e.g., whether a complaint can be resolved or not).

Test ID	Test Case Description	Input Data	Expected Output	Type	Result
BVA-C-001	Test logging of a valid complaint	Date: 2024-08-15, Description: Issue with AC	Complaint logged successfully	BVA (Valid Boolean Input)	Passed
BVA-C-002	Test logging with missing description	Date: 2024-08-15, Description: NULL	Error: Description is required	BVA (Invalid Boolean Input)	Passed

Table 32: BVA based on Boolean values in complaint

Conclusion of Test Cases

The above tests cover various scenarios using ECP and BVA separately:

- **ECP:** Focuses on valid/invalid inputs and Boolean values.
- **BVA:** Concentrates on boundary values with single and multiple variables, as well as Boolean values.

A comprehensive database system is essential for Dickwella Beach Hotel and Resort to optimize its operations. This system effectively fulfils the hotel's needs by managing room bookings, guest details, reservations, services, billing, and customer complaints.

Thorough testing using Equivalence Class Partitioning and Boundary Value Analysis verified the system's performance under diverse conditions, identifying and resolving any issues. The database system has proven capable of handling the hotel's intricate operations, from initial booking to final payment. This robust solution will significantly improve efficiency and satisfaction for both staff and guests.

By structuring tests this way, we ensure comprehensive coverage of all critical aspects of the hotel management system's database. Each test case has been **validated**, and all test results **passed**, “confirming the system's behavior aligns with the specified requirements.”

Task 4

Task 4.1.

- a. Technical user documentation for developed database system explaining how to use the database system.

Note: The documentation added within the folder. Please check the PDF formatted documentation.

Task 4.2

- a. Produce a comprehensive design for the fully-functional database system, which includes user interface design.

1. Login interface

Purpose: The LoginForm is the entry point to the hotel management system. It is designed to authenticate users before they can access the main dashboard.



Figure 51:login interface

2. Main Dashboard

Purpose: The MainDashboardForm serves as the central hub of the hotel management system. It provides access to all management features like guest management, room management, and others.

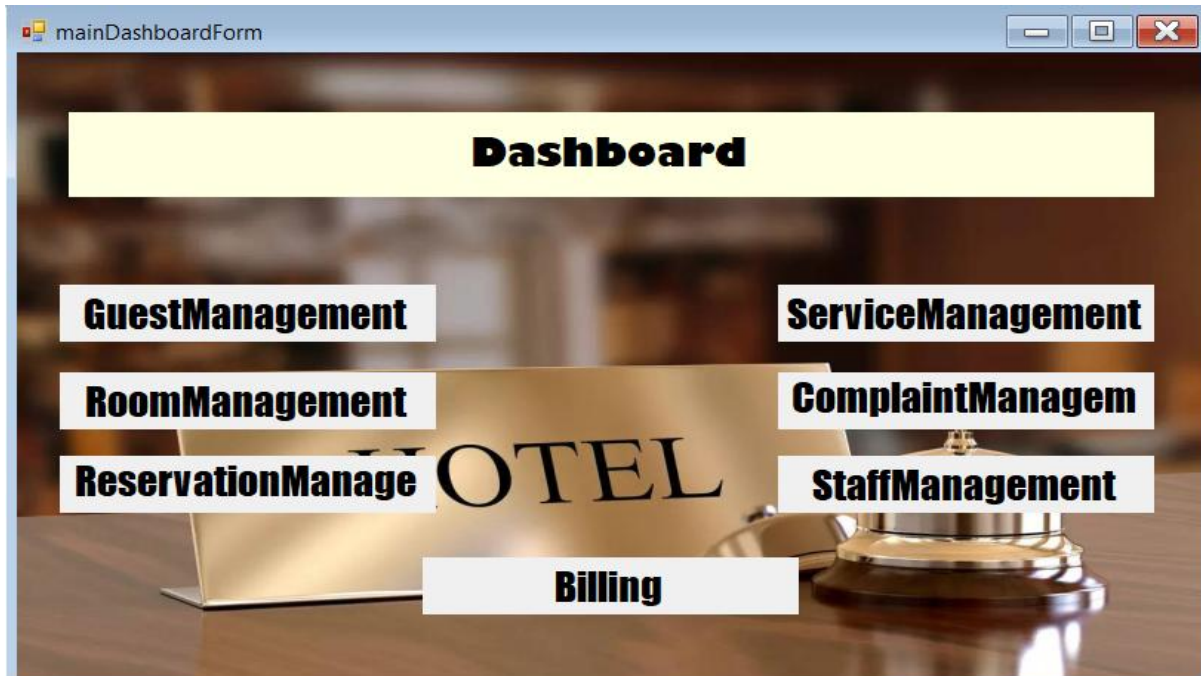


Figure 52:main dashboard

3. Guest Management

Purpose: The GuestManagementForm allows the hotel staff to add, update, or delete guest information. It also displays a list of all current guests.



The screenshot shows a window titled 'guestManagementForm'. The main content area has a background image of a hotel lobby. Overlaid on this is a form titled 'Guest Management'. The form contains several input fields and buttons. On the left, there are labels for 'Guest ID', 'Name', 'Occupancy', 'Date of Birth', 'Address', 'Age', and 'Contact number', each followed by a text input field. On the right, there are four buttons: 'Create', 'Update', 'Delete', and 'Search'.

Figure 53:Guest management UI

4. Room Management

Purpose: The RoomManagementForm is used to manage the details of the hotel's rooms, such as room type, status, charges, and other attributes.



The screenshot shows a web application window titled "RoomManagementForm". Inside, there is a yellow header bar with the text "Room Management". Below this, there are six input fields on the left, each with a label: "Room ID", "Facility", "Type", "Charge", "Capacity", and "Status". To the right of these fields are four buttons: "Create", "Update", "Delete", and "Search". The background of the form is a blurred image of a hotel interior.

Figure 54:room management UI

5. Reservation Management

Purpose: The ReservationManagementForm handles all tasks related to managing guest reservations, including creating new reservations and viewing or editing existing ones.



The screenshot shows a web application window titled "ReservationManagementForm". Inside, there is a yellow header bar with the text "Reservation Management". Below this, there are five input fields on the left, each with a label: "Reservation ID", "Start Date", "End Date", "Payment status", and "No of guest". To the right of these fields are four buttons: "Create", "Update", "Delete", and "Search". The background of the form is a blurred image of a hotel interior.

Figure 55:reservation management UI

6. Additional Service Management

Purpose: The ServiceManagementForm manages additional services provided by the hotel, such as spa, laundry, or room service. Staff can add new services, update existing ones, or remove services.

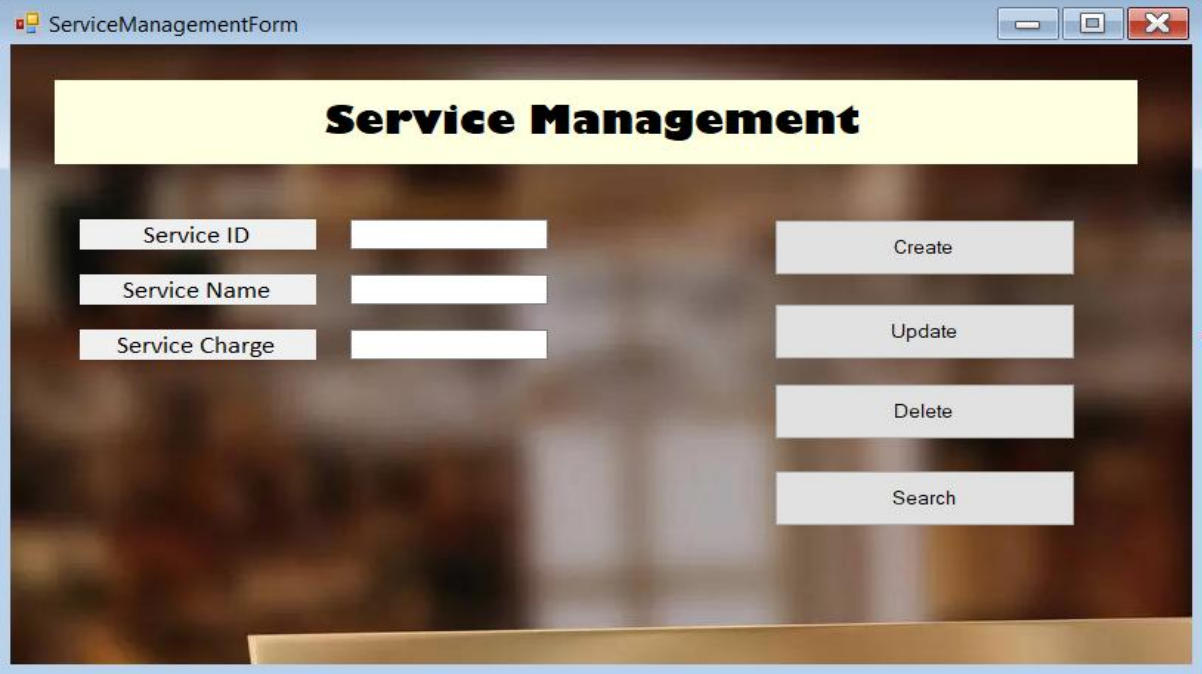


Figure 56:Additional services management UI

7. Employee Management

Purpose: The StaffManagementForm is used to manage the details of hotel staff, including their roles and room assignments.



Figure 57:employee management UI

8. Complaint Management

Purpose: The ComplaintManagementForm manages guest complaints, allowing staff to record, update, and resolve complaints.



The screenshot shows a web application window titled "ComplaintManagementForm". Inside, there is a yellow header bar with the text "Complaint Management". Below this, there are three input fields on the left: "Complaint ID", "Date", and "Description". To the right of these fields are four buttons: "Create", "Update", "Delete", and "Search". The background of the form is a blurred image of a person.

Figure 58: Complaint management UI

9. Billing

Purpose: The BillingForm is used to calculate and generate the final bill for a guest's stay, including room charges, services, and any additional fees.



The screenshot shows a web application window titled "BillingForm". Inside, there is a yellow header bar with the text "Billing". Below this, there are four input fields on the left: "Bill ID", "Total amount", "Deducted amount", and "Final amount". To the right of these fields are four buttons: "Create", "Update", "Delete", and "Search". The background of the form is a blurred image of a person.

Figure 59: Billing Interface

Conclusion

In conclusion, the development of the hotel management system, as outlined in this project, demonstrates a comprehensive approach to managing various aspects of hotel operations through a structured and user-friendly interface. The system integrates several key functionalities, including user authentication, guest management, room management, reservation management, service management, staff management, complaint handling, and billing. Each module was carefully designed to streamline processes, reduce manual errors, and enhance the overall efficiency of hotel operations.

The implementation of the system using C# and Windows Forms ensures that it is robust and scalable, allowing for easy modifications and future enhancements as the hotel's needs evolve. By utilizing a database-driven approach, the system maintains data integrity and provides quick access to information, which is crucial for effective decision-making.

Throughout this project, careful consideration was given to the user experience, with intuitive interfaces and straightforward navigation paths that enable staff to perform their duties with minimal training. The system's modular design also allows for easy integration with other hotel systems, offering the potential for further expansion.

Overall, this project serves as a practical application of software development principles in a real-world context, showcasing the importance of combining technical knowledge with an understanding of user requirements. The resulting system is a valuable tool that not only meets the current needs of the hotel but also lays a strong foundation for future technological advancements in hotel management.